Alcohol Availability, Prenatal Conditions, and Long-Term Economic Outcomes

J. Peter Nilsson

Stockholm University, Uppsala University, and Institute for Evaluation of Labour Market and Education Policy

This study examines how a policy that sharply increased alcohol availability during 8.5 months affected the labor productivity of those exposed to it in utero. Compared to the surrounding cohorts, the prenatally exposed children have substantially worse labor market and educational outcomes and lower cognitive and noncognitive ability. Effects on earnings are found throughout the distribution but are largest below the median. Males are more affected than females, consistent with growing evidence that boys are less resilient to early environmental insults. The long-term effects seem primarily driven by changes in prenatal health rather than changes in the childhood environment.

I. Introduction

Starting with Jones and Smith (1973), a vast literature has documented negative correlations between heavy prenatal alcohol exposure and infant health, early childhood cognitive ability, and behavior. Yet very little is known about the effects on adult outcomes. Moreover, the earlier (al-

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most exclusively cross-sectional) studies prohibit clear conclusions since alcohol use during pregnancy is likely correlated with many unobserved factors that also influence child outcomes.[1] For example, alcohol use is associated with a higher risk of unplanned pregnancy; unplanned pregnancies, in turn, are associated with family instability, maternal labor supply, and lower human capital investments in children (see, e.g., Joyce, Kaestner, and Korenman 2000; Kaestner and Joyce 2001; Naimi et al. 2003; Grossman and Markowitz 2005; Nuevo-Chiquero 2014). Hence, unless the effect of alcohol on selection into pregnancy is fully accounted for, the estimated effects of prenatal alcohol exposure on child outcomes are likely to be biased.

The primary innovation of this paper is to examine the long-run impact of an alcohol policy experiment on the children exposed to it in utero. The policy temporarily and sharply increased access to strong beer in certain regions of Sweden during the late 1960s, particularly, and inadvertently, for people under the age of 21.[2] Administrative data covering all children born in Sweden between 1964 and 1972 allow me to follow the exposed children for more than 30 years and provide a detailed account of the policy's impact on their labor market outcomes as adults.

The results section of the paper begins by documenting the immediate impact of the 8.5-month-long policy experiment. Using several indicators of alcohol consumption, I find large effects on young women's alcohol use. At the same time, there was a deterioration of health-related outcomes of children of young mothers in early gestation. The increase in alcohol availability also changed the parental composition of children of young mothers conceived during the policy, as reflected in family background characteristics.

I then assess the long-run effects of the policy and find a substantial impact on the labor market outcomes of children conceived before and exposed to the policy in utero. In particular, in comparison to the surrounding cohorts, those with the longest prenatal exposure to the policy who were born to mothers under the age of 21, on average, have around 20 percent lower earnings and are more likely to have no earnings and collect welfare payments as adults. Negative effects are evident throughout much of the earnings distribution, but the mean effects mask a particularly strong impact below the median. At the bottom of the distribution, the lower earnings seem to stem from lower labor supply (unemployment). Further up in the distribution, wage effects play a more important role.

The exposed cohort has significantly lower high school completion and college attendance rates, precisely the educational margin that can

[1] Systematic reviews have found no convincing evidence of adverse effects of consumption that are common at the population level (Henderson, Gray, and Brockelhurst 2007; Henderson, Kesmodel, and Gray 2007). See also online app. B.

[2] Strong beer is restricted to an alcohol content of 5.6 percent by volume.

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be expected to have a strong impact on labor market outcomes. Moreover, using military enlistment data, I find that the probability of having low cognitive and noncognitive ability (extraversion, taking responsibility, independence, etc.) increased by 27 and 16 percent, respectively. Low noncognitive ability is strongly linked to low employment rates. However, as reflected in the relatively muted impact on disposable income, public transfers dramatically reduced the pecuniary effects of the policy. This suggests that the large effects on earnings, at least in part, could also be the result of reduced labor supply incentives or reduced labor demand for low-skilled workers induced by the generous welfare system.

The results are robust to several specification checks, including comparisons with children born in neighboring regions, children born just before or conceived just after the policy, and the exposed children's unexposed siblings. These findings provide additional evidence indicating that the estimated impact of the policy is not driven by unobserved factors coinciding with the policy or by general family-specific factors.

To my knowledge, no previous study has estimated the effects of prenatal exposure to alcohol or alcohol policies on adult earnings or wages, and as noted above, the vast majority of previous studies are correlational.[3] This paper also contributes to the broader and rapidly growing literature interested in the early-life determinants of medium- and long-term outcomes in several ways.

First, I make use of unusually detailed and high-quality data to document the effects of the policy on a wide range of adult outcomes. Previous studies on fetal insults are usually restricted to examining effects on school outcomes (Almond, Edlund, and Palme 2009; Field, Robles, and Torero 2009; Sanders 2012; Almond, Mazumder, and van Ewijk 2015). Expected earnings effects are then typically imputed using estimates of the effects of education onto earnings or wages. There are, however, many reasons why such imputations are likely to understate the true effect. Moreover, Bleakley (2010) underscores that it is not obvious that school outcome is a sufficient statistic for measuring the impact on lifetime income and that providing direct evidence on earnings is important.

Second, early work on the long-term effects of in utero conditions focused on rare or extreme events.[4] I provide evidence of substantial long-

[3] Notable exceptions are the studies by Fertig and Watson (2009), Zhang (2010), and Barreca and Paige (2015), who examine effects of alcohol policies on birth outcomes. Given the crudeness of birth outcomes such as birth weight as measures of health, it is crucial to also document the impact of alcohol policies on actual later-life outcomes.

[4] Important early examples of effects of extreme events include the influential epidemiological studies on long-term effects of the famine during the Dutch Hunger Winter and the Great Famine (Chen and Zhou 2007) and the work on the impact of the Spanish influenza pandemic on subsequent socioeconomic and health outcomes of those in utero during the peak of the epidemic (Almond 2006).

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term effects of common and less extreme conditions, potentially more easily influenced by policy makers.

Third, in line with previous work, I find a lower sex ratio at birth in the most exposed cohorts, which suggests that males were more likely to be spontaneously aborted or born prematurely.[5] This study adds to this research by documenting that prenatal exposure to the policy affected boys more than girls also in the long run.

Fourth, most previous studies assessing effects of prenatal shocks on school outcomes use data from developing countries and generally find that reinforcing postnatal investment responses explains part of the impact of early health on academic achievement. It is not clear to what extent these results generalize to labor market outcomes or to developed countries. In this setting, I find no clear indication that parents have systematically reinforced or compensated for the changes in initial endowments.

Finally, I provide evidence of the effects on completed fertility in the parental cohort (F0), near-completed fertility in the exposed cohort (F1), and effects on health at birth in the subsequent generation (F2). To my knowledge, no previous studies on prenatal alcohol exposure, alcohol policy, or other prenatal shocks have been able do this for both F1 parents. Despite the large negative effect on adult outcomes, I find no indication that parental prenatal exposure to the policy has affected neonatal health in the subsequent generation.

The rest of the paper is organized as follows: Section II presents the conceptual framework, and Section III describes the details on the policy experiment and documents the immediate effects. Section IV describes the data and the empirical strategy. Section V presents the effects on early health, selection, labor market outcomes, mediating mechanisms, and robustness checks. Section VI presents evidence on the role of postnatal responsive investments and the impact on health in the next generation. Section VII summarizes and concludes the paper.

II. Conceptual Framework: Critical Periods and Human Capital Capacity Formation

The appreciation of critical early windows (Davison and Dobbing 1968) during which environmental perturbations could have lifelong consequences has spurred a large literature investigating the early-life origins of adult disease (Barker 1998; Gluckman and Hanson 2004). Economists have added to this literature by identifying the role of early-life circum-

[5] See Trivers and Willard (1973), Wells (2000), Norberg (2004), Almond and Edlund (2007), Almond and Mazumder (2011), Sanders and Stoecker (2011), and Barreca and Page (2015).

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stances on nonhealth outcomes and by formalizing a capacity formation model accounting for the varying returns on human capital investments over the child's life cycle (Cunha and Heckman 2007). Cunha and Heckman formalize and stress the importance of recognizing the role of sensitive and critical periods for optimal human capital investments.

Following Cunha and Heckman's framework, I now describe the technology forming human capital accumulation over the child's life cycle to provide intuition for the empirical results. Assume that an agent at conception has the initial condition θ_1. Let θ_P denote parental characteristics (e.g., cognitive skills, education). At each stage t, let θ_t denote the vector of skill stock. The technology of production of skill after t periods is

$$\theta_{t+1} = f_t(\theta_P, \theta_t, I_t) \tag{1}$$

for periods of childhood $t = 1, 2, \ldots, T$.

The technology in (1) is written in recursive form. Substituting in (1) for $\theta_t, \theta_{t-1}, \ldots$, it is possible to rewrite the stock of skills at stage $t + 1$, θ_{t+1}, as a function of the initial conditions, parental skills, and all of the past investments:

$$\theta_{t+1} = m_t(\theta_P, \theta_1, I_1, \ldots, I_t), \quad t = 1, \ldots, T. \tag{2}$$

Cunha and Heckman (2007) highlight two important features in the capacity formation. First, "dynamic complementarities" may arise when $\partial^2 f_t(\theta_P, \theta_t, I_t)/\partial\theta_t \partial I_t > 0$, that is, when stocks of skills acquired by the end of period $t - 1$ (θ_t) make investment in period t (I_t) more productive. Second, "self-productivity" arises when $\partial f_t(\theta_P, \theta_t, I_t)/\partial\theta_t > 0$, that is, when higher stocks of skills in one period create higher stocks of skills in the next period.

For simplicity suppose that there are only two childhood periods, $T = 2$, in the child's life cycle, one prenatal ($t = 1$) and one postnatal stage ($t = 2$), and that the human capital production function takes the following constant elasticity of substitution functional form:

$$h = m_2\left(\theta_P, \theta_1, \left[\gamma \tilde{I}_1^{\phi} + (1 - \gamma) I_2^{\phi}\right]^{1/\phi}\right), \tag{3}$$

where $\gamma \in [0, 1]$ and $\phi \in (-\infty, 1]$. The share parameter (γ) is here a skill multiplier, and $\tilde{I}_1 = I_1 + \mu$, where μ is an exogenous shock occurring in period 1. The elasticity of substitution, $1/(1 - \phi)$, is a measure of the substitutability of $\tilde{I}_1$ and I_2; ϕ represents the degree of complementarity/substitutability and determines how easy it is to compensate for low levels of investments (or a negative μ) in the prenatal stage during the postnatal stage.

When ϕ is small, it is difficult to compensate for low levels of prenatal investments ($\tilde{I}_1$) during the postnatal period (I_2). When $\phi = 1$, that is, $\tilde{I}_1$

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and I_2 are perfect substitutes, the timing of investments (pre- or postnatal period) is irrelevant for the level of human capital in adulthood. In the other extreme case, $\phi \to -\infty$, it is impossible to compensate for low prenatal investments in the postnatal period.

Early pregnancy is a critical/highly sensitive period, so in the context of this paper, $\phi < 0$ is likely the empirically relevant case. In this case, even a small shock may result in large negative outcomes in the long run. The μ effect is carried over to the following period, and the combined effects of self-productivity and dynamic complementarities magnify its impact on human capital stock over time, which could have lasting lifetime consequences that are difficult to remediate at later ages.

An interesting finding shown below is that the negative early prenatal shock examined in this paper generally has a larger effect for boys than for girls. Previous studies have often found that girls are more resilient to poor early conditions than boys, as reflected by, for example, higher prematurity rates, higher morbidity, and mortality in early life (Lindström 1999; Wells 2000).[6] One way to capture these findings in (3) is to allow ϕ to vary by gender, with $\phi_{\text{boys}} < \phi_{\text{girls}}$.[7] Under this parameterization, the elasticity of substitution across the prenatal and postnatal periods is lower for boys than for girls, implying that it is more difficult to remedy an early shock for boys than for girls. Via dynamic complementarities and self-productivity of skills an early shock could hence result in larger effects in males than in females.[8]

The Cunha-Heckman framework above highlights how sensitive and critical periods in development affect the technical possibilities of reme-

[6] Male sex is a significant risk factor for adverse neonatal outcomes. For example, among newborns, the secondary sex ratio (share of males at birth) among full-term babies is 0.505. Among extremely premature and very premature births (gestational age 24–34 weeks), the sex ratio is constant at 0.55. For premature babies (weeks 35–37) the share of males decreases linearly by gestational week down to around 0.525. Until recently, premature babies had a hard time surviving until reproductive age, and despite substantial public medical care investments, prematurity is still associated with adverse adult outcomes (source: own calculations using data from the Swedish Medical Birth Register).

[7] Thanks to the editor James Heckman for suggesting this formulation.

[8] It should also be noted that postnatal responsive investments could further increase these differences. Gender differences in the long-term effects could, e.g., also emerge because of gender discrimination in parental investments. Almond et al. (2012) show that in societies (China) with strong son preferences and less than perfect complementarity between investment periods, the effects of a similarly sized prenatal shock can yield larger effects in boys than in girls even in the absence of any biological differences. While this alternative explanation for the differences in long-term effects documented below is difficult to completely rule out, it should be stressed that (1) there are no indications of strong son preferences in Sweden during the sample period (see, e.g., Andersson et al. 2006); (2) it is hard to reconcile the effects on the sex ratio shown below being driven by gender discrimination in postnatal investments; and (3) the maternal fixed-effect analysis below suggests no major indication that (postnatal) responsive investments account for the effects on long-term outcomes. Together, this suggests that a model with gender differences in effects of prenatal shocks is more appropriate than a model assuming no such differences in this context.

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diation. Others, for example, Heckman and Mosso (2014), develop a formal model of child development showing how the capacity formation technology in (3) affects the optimal investment profiles in an overlapping generations economy and how parents respond to low endowments.

Online appendix B provides a review of studies on the direct and indirect (e.g., reduced placenta functioning) physiological mechanisms through which prenatal alcohol exposure has been suggested to influence early development.[9]

III. The Immediate Impact of the Policy Experiment

Alcohol sales in Sweden are strictly regulated by an off-premises retail monopoly (*Systembolaget*).[10] The current retail system has been in effect since 1955. Since then the consumption pattern has changed radically. Sweden traditionally belonged to the "spirits-drinking countries." In 1966 spirits accounted for 56 percent of total sales, while the dominant beverages today are strong beer (29 percent) and wine (44 percent; Swedish National Institute of Public Health 2005).

The changing patterns are partly due to policies designed to encourage substitution from spirits to beverages with lower alcohol content.[11] The strong beer policy experiment between November 1967 through July 1968 in Göteborg-och Bohuslän and Värmland regions (jointly 12 percent of the population) is an example of an intervention with this intent.[12] During the policy, off-premises sales of strong beer were allowed in regular grocery stores as compared to only in the Systembolaget stores prior to and after the policy.[13] The intention was to terminate and evaluate the policy at the end of 1968, but soon after implementation, reports

[9] In addition, besides alcohol effects, additional negative effects of the increase in alcohol availability may come through changes in behaviors that are complements to alcohol consumption, such as smoking (Dee 1999), which are also negatively associated with prenatal development. Attempts to assess the effects of alcohol use in comparison with the use of other drugs have, however, suggested that prenatal alcohol exposure may result in broader and more long-lasting effects compared to other drugs; see, e.g., Day and Richardson (1994).

[10] The only alcoholic beverages permitted in regular grocery stores are those containing less than 3.5 percent alcohol by volume.

[11] See Room (2002) for a comprehensive review of Nordic alcohol policies after 1950.

[12] The setup and results of the experiment are described in SOU 1971:77, on which this section draws. In the report no motivation is given as to why the two regions were selected from the pool of 24 regions. I use "sales" and "consumption" interchangeably below. Sales are strongly correlated with alcohol consumption for both men and women.

[13] At the end of 1968, 1,530 retail outlets were licensed for sales of beer (during the experiment also strong beer) in Göteborg och Bohuslän region as compared to the 26 Systembolaget stores in operation prior to and after the experiment. Anyone entitled to sell or serve beer was allowed to buy strong beer directly from a Swedish brewery or wholesal-

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of a sharp increase in alcohol consumption in the experimental regions, especially among young people, were received. This caused the implementing authority, the Alcohol Policy Commission (APU), to propose an interruption, and on July 15, 1968, the policy experiment was discontinued prematurely.

A. Impact on Alcohol Sales

There are excellent opportunities for evaluating the impact of the policy experiment on wine, spirits, and strong beer sales. Systembolaget kept exact records of on- and off-premise sales in each region prior to, during, and after the experiment.[14] The top panel of figure 1 shows that prior to the policy, strong beer consumption in the treated regions and the rest of Sweden followed each other almost perfectly. During the policy, consumption increased by 1,000 percent and then declined again. Note that strong beer sales remained at an elevated level after the policy ended. This indicates that a short-term experiment could have long-term effects on consumption (SOU 1971:77).[15]

The bottom panel of figure 1 shows that the policy's intention to reduce spirits consumption failed. Changes in liquor (and wine) sales were small and did not nearly compensate for the substantial increases in sales of strong beer. However, it is highly likely that the increased sales of strong beer led to a decline in the sales of medium beer, as these products are arguably closer substitutes.[16] Medium beer was sold in grocery stores before, during, and after the policy with an age limit of 16, but unfortunately, there are no regional data on sales. Using aggregate monthly data, the APU calculated that the reduction should have been around 10 mil-

ers. Breweries and wholesalers were obligated to reported the amount of strong beer shipped to retailers.

[14] On-premise sales in relationship to off-premise sales were very low at the time of the policy: 13 percent in Göteborg and 7 percent in Värmland. The national age limit for on-premise sales was 16 until July 1972 and then 18.

[15] There is a clear connection between consumption and population density. Per capita consumption was highest in Göteborg (684,626 inhabitants) followed by Karlstad (53,208 inhabitants) and Uddevalla (36,480 inhabitants). The reason for this pattern is probably greater availability in urban areas. Another reason may be that people in rural areas bought strong beer when visiting the cities. However, it is also likely that some cross-border shopping for beer occurred during the experiment at least by consumers in the neighboring regions. This suggests that an experiment including the whole country would have generated a smaller increase in consumption per capita. The extent of cross-border shopping is unknown, but it seems unlikely that it had any major influence on total sales. The reason is that while availability increased, prices (if anything) increased slightly during the policy compared to the Systembolaget prices (SOU 1971:77). In the empirical section, I also check whether the policy affected children born in the neighboring regions.

[16] Medium beer contains between 3.5 and 4.5 percent alcohol by volume.

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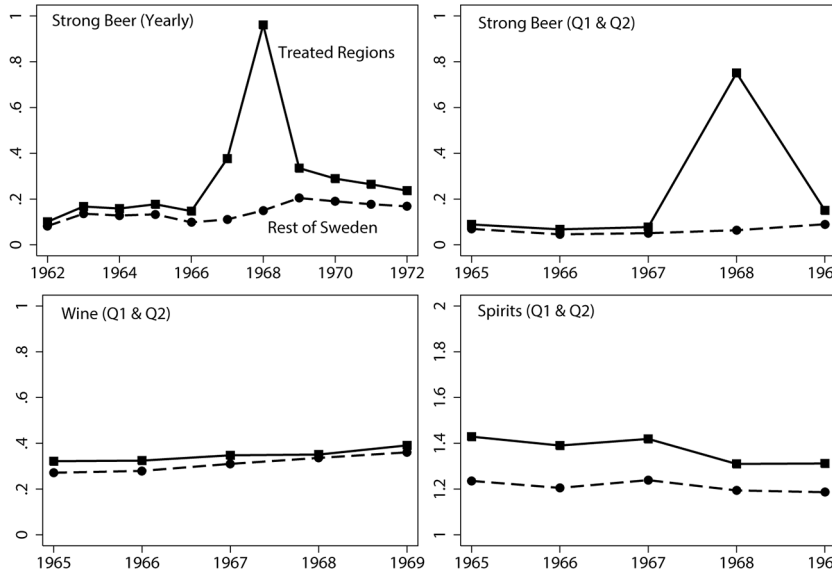


FIG. 1.—Strong beer, spirits, and wine sales, in liters of 100 percent alcohol per capita. Sources: Yearly data from Statistics Sweden (1962–72); quarterly data from SOU (1971:77). Data are not available for the quarters individually.

lion liters overall.[17] This should be compared with the extra 11.8 million liters of strong beer consumed in the treatment regions.

On the basis of these calculations, the average increase in the treatment regions in terms of liters of 100 percent alcohol has previously been estimated to be around 5 percent (SOU 1971:77). Relative to the national 1967 baseline of 6.3 liters of 100 percent alcohol consumption per capita, this corresponds to a net increase of 13 33-centiliter bottles of strong beer per capita (or about three extra binge drinking occasions) during the 8.5-month experiment. This back-of-the-envelope calculation does not consider that alcohol consumption generally follows a bimodal pattern over the year (peaks in December and in the summer), nor that, for example, young and old may have responded differentially to the increase in availability of strong beer.

Alcohol consumption data are not available for subpopulations, which hinders estimations of the exact magnitude of the changes in consump-

[17] The national consumption of medium beer increased by only 14 percent during the first 6 months of 1968, compared with an increase of 25 percent for the first three quarters of 1967 and 35 percent during the fourth quarter of 1968. This suggests that the experiment led to a reduction in the increase of medium beer sales of about 10 percentage points and that strong beer to some extent replaced medium beer. During the first 6 months of 1967, 91 million liters of medium beer were sold (SOU 1971:77).

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tion among subgroups.[18] At least we know from a nationwide survey among 15–25-year-olds conducted in the spring/summer of 1968 that beer consumption was 44 percent higher among young people than in the population as a whole.[19] Importantly, the survey also reveals that in 1968, 90 percent of the females reported that their alcohol debut occurred before turning 21 and that the reported abstainer rates among young women were low.[20] The following sections provide evidence on differential effects across subgroups using alternative alcohol consumption indicators.

B. Evidence from a Survey

In the late spring of 1968 the APU surveyed the local child welfare commissions (*barnavårdsnämnder*), the temperance commissions (*nykterhetsnämnder*), the local education authorities, and the police authorities in the experimental regions regarding their experiences of the policy hitherto. The main conclusion from this survey was that alcohol consumption had increased. Using the original response sheets, I linked the responses to the municipality of birth for the children born in the treatment regions.[21] More than 90 percent of the children were born in municipalities where both the child welfare commission and the temperance commission reported that conditions had deteriorated as a consequence of the policy experiment.[22] The municipalities that had not experienced any notable changes are predominantly small rural municipalities.

According to the APU report, the police authorities underscored that alcohol consumption had increased, particularly among young people. The main nuisances reported were an increased level of disorderly conduct and littering in connection with an immense consumption of strong

[18] Considerable effort was made to find alcohol consumption data for subgroups. The lack of data from the experiment on subgroups reflects the unexpected consequences of the policy. An illustrative example of this can be found in a reservation statement in the final report from the policy made by Commissioner Rune Hermansson, who was part of the commission that designed and evaluated the policy. In his reservation Hermansson states that "although there is no doubt that total alcohol consumption increased substantially because of the experiment, we do not know whether this was just a temporary increase or would last if the policy was permanent." He urges that more data should be collected but continues, "other members of the APU have shown no interest in collecting such complementary data" (SOU 1971:77, 8:43; my translation).

[19] See SOU 1971:77. Unfortunately, the raw data from this survey are not available for further analysis.

[20] In the highest, middle, and lowest social strata, 2, 8, and 10 percent of the young women (aged between 17 and 25) reported no alcohol consumption in 1968 (SOU 1971:77).

[21] Records are kept at the Swedish National Archives, SE/RA/420144/420144.05/H2/50-54.

[22] The school authorities' responses are ignored since many of them report having contact only with kids aged 7–12.

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beer. An increase in DUIs was also noted. Urban areas seem to have been more affected than rural areas (SOU 1971:77).

C. A Legislative Loophole

One important explanation for the particularly strong effects on consumption among young people could be that they experienced the largest increase in the availability of alcohol during the policy.[23] The age limit in Systembolaget stores was set to 21, and prior to the experiment this was the only off-premise place where strong beer could be bought. The minimum purchasing age for beer in regular grocery stores during the policy was 16, although the enforcement of this law was weak (SOU 1974:91).

Hence, in line with the intention, the majority of the population in the experiment regions experienced an increase in availability of a relatively low alcohol content beverage. But this resulted in only a small reduction in consumption of spirits and wine. On the contrary, for those without the possibility of buying alcohol at Systembolaget, the policy sharply increased availability of a higher alcohol content beverage. This discrepancy in changes in availability for those below and above age 21 was either not realized by the APU or ignored in their evaluation. However, the loophole was reported by the dominant newspaper in the region already on the second day of the policy experiment.[24] The age-specific differences in changes in alcohol availability provide one plausible explanation for the reported differences in the effects of the policy.

D. Impact on Arrests for Drunkenness

In contrast to the police reports, the original evaluation by the implementing authority APU found no clear effect of the experiment on arrests for drunkenness (SOU 1971:77), although they noted that during the period alcohol consumption was increasing while arrests were going down. I reexamined their conclusion using consistently reported city-by-quarter data on arrests for drunkenness for the period 1964–70 in 97 cities.

Panel A in Appendix table A1 shows difference-in-differences (DD) estimates suggesting a significant increase in arrest rates in Q4:1967 (15 per-

[23] For the effects of alcohol availability on consumption patterns in general, see, e.g., O'Malley and Wagenaar (1991) for US evidence and Norström and Skog (2005) for Sweden. Several previous studies focusing on young people have found responsiveness to policies pertaining to availability, such as the minimum legal drinking age (MLDA) laws; see, e.g., Moore and Cook (1995). Cook and Moore (2001) found that there is very clear evidence that the age-specific legality of drinking is a key determinant of the age pattern of drinking, and particularly of binge drinking.

[24] See *Göteborgsposten*, November 2, 1967. Also see SFS (1961:159, 1967:213) for the laws in effect during the policy.

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cent) and Q1:1968 (13 percent) and a nonsignificant (5 percent) increase in the last quarter of the policy (Q2:1968). Yearly sex- and age-specific (above/below age 21) arrest data are available for the three largest cities (Stockholm [control], Göteborg [treated], and Malmö [control]) and in aggregate form for the rest of the major cities between 1964 and 1968. Panel B shows that those above age 21 were not affected, while arrests among young women (men) increased by 20 percent (14 percent).

Besides the limited sample sizes, it is not unproblematic to use arrest data to proxy for alcohol consumption since the number of arrests also depends on the enforcement of the law. If the police became more lenient (stricter) during the policy, then changes in arrests understate (overstate) the effects of the policy on alcohol consumption. However, as is shown below, the timing of the increase in arrests among young people coincides perfectly with an increased negative selection into pregnancy and with a substantial increase in unstable family formations among young people. These indicators for alcohol consumption, together with the effects on alcohol sales, the reports from the local authorities, the legislative loophole, and the newspaper reports, corroborate the reports of a sharp increase in alcohol consumption among young people in the experimental regions. Jointly they provide an important prior: children born to mothers under age 21 are likely to have been most affected by the policy.

IV. Empirical Strategy

Considering the low awareness of the risks associated with alcohol consumption during pregnancy during the policy and the indications of a sharp increase in alcohol consumption among young people, clearly the long-run outcomes of children exposed to the policy in utero may have been affected.[25] Although there is plenty of evidence from animal experiments of negative effects of prenatal alcohol exposure on offspring outcomes, human studies are almost exclusively correlational. It is therefore difficult to predict the extent of the effects of the policy on the basis of previous work. This section describes how the policy experiment can be used to circumvent many of the methodological problems of previous studies.

[25] The first general warning about the association between alcohol consumption during pregnancy and birth defects was issued by the US Surgeon General and the Swedish National Board of Health and Welfare in the early 1980s, as a response to the increasing evidence gathered from the 1970s and on (see App. fig. A1 and online fig. B1). Still today there exists some variation in recommendations across countries; e.g., while the United States, Sweden, and France suggest that women abstain from alcohol when planning to become pregnant and throughout pregnancy, in the United Kingdom, pregnant women are recommended to abstain during the first 3 months of pregnancy. Still a significant share of women drink at least up until pregnancy recognition.

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To quantify the extent of the long-term effects on labor productivity, I use the LOUISE database assembled by Statistics Sweden covering all individuals in the age range 16–65 living or working in Sweden between 1990 and 2004. The LOUISE data are register based, and apart from information on year and month of birth, gender, and region of birth, the database also contains detailed information on educational attainments, labor market outcomes, and welfare payments received during the observation period. Each individual in the data is linked to his or her biological parents using the "multigenerational" register.

In the main analysis, all firstborn individuals alive in 2000 and born in Sweden between 1964 and 1972 are retained.[26] The children born in the five regions neighboring the experimental regions are at first excluded in order to avoid diluting the estimates as a result of potential spillover effects from the policy. As the experiment was implemented at the regional level, I use panel data for regions (in total 24 regions). However, to allow for the age-specific differences of the policy shift on availability and consumption, the sample is further partitioned with respect to the age of the mother at delivery (below/above age 21).

On the basis of exposure to the policy, children born in the treatment regions are divided into five groups: (1) those born prior to the initiation of the experiment and, hence, exposed only after birth; (2) those exposed to the experiment from late pregnancy but conceived before the experiment started; (3) those exposed to the experiment from early pregnancy but conceived before the experiment started; (4) those prenatally exposed but conceived during the course of the experiment; and (5) those who were conceived after the end of the experiment and thus not exposed either before or after birth.

In the baseline estimations, I focus on children belonging to group 3 for two reasons. First, it seems reasonable to assume that the policy did not affect the timing of conception for this group of children. By focusing on children conceived prior to the start of the experiment, biased estimates of the relationship of interest due to any compositional effects caused by the experiment are effectively avoided. This is important, as several studies have found an association between alcohol consumption and risky behavior/unplanned pregnancies among young people (Kaestner and Joyce 2001; Carpenter 2005; Grossman and Markowitz 2005; Carpenter and Dobkin 2009). Fertig and Watson (2009) find that MLDA laws affect infant health mainly through their effect on family

[26] Firstborns are first of all singled out because of the assumption that people without previous children are more likely to react to a temporary increase in alcohol availability. In addition, given the focus on mothers under age 21, adding higher-order birth children will have only a marginal effect on the size of the treatment group since few women give birth to two children before age 21.

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composition.[27] In Section V.B, I provide evidence in support of this notion. Second, the group 3 children were exposed for the longest duration and from early pregnancy. However, the impacts on children in late gestation and the four other exposure groups are considered in the analysis as well.

Table 1 displays the estimated date of conception, duration of exposure, gestational age at the start of the experiment, and trimester under exposure based on month of birth. The main exposure cohort (conceived between July and October 1967) is group 3. The main specification used is the following baseline difference-in-difference-in-differences (DDD) model (Gruber 1994):

$$Y_{r,t,m<21} = \alpha_0 + \beta_3 \text{EXPOSURE}_{3,r,t,m<21} + \eta_{r,t} + \eta_{r,m<21} + \eta_{t,m<21} + \varepsilon_{r,t,m<21}, \quad (4)$$

which is estimated by ordinary least squares on data aggregated to year-by-month (t), age of mother (below/above 21) ($m<21$), and region of birth (r).[28] The term $Y_{r,t,m<21}$ refers to the outcomes of interest (e.g., average earnings, share zero income, share on welfare); $\text{EXPOSURE}_{3,r,t,m<21}$ is equal to one if the child is born to a mother under the age of 21 at delivery in the treatment regions and conceived between July and October 1967, and otherwise zero; β_3 is the parameter of interest reflecting the impact of the policy on the children who were exposed to the policy from early pregnancy and were born to a young mother.

The term $\eta_{r,t}$ is period by region of birth specific effects, accounting for region by period of birth specific factors influencing adult outcomes;[29] $\eta_{r,m<21}$ are region by maternal age at birth effects, accounting for fixed region-specific (and general) differences between children born to mothers above/below age 21 and, for example, regional differences in the

[27] Fertig and Watson (2009) find fairly small effects on birth outcomes, although the authors also suggest that this could be due to the fact that the MLDA had only a modest effect on consumption. Additionally, birth outcomes such as birth weight may not be an ideal measure when it comes to alcohol exposure since birth weight is mainly determined in the later stages of the pregnancy. Since drinking during pregnancy typically decreases sharply with gestation, it is notable that Fertig and Watson find significantly negative effects on birth weight from the MLDA changes. This could indicate that the full effect on fetal development from the MLDA policies is larger than what the effects on birth outcomes reveal.

[28] The aggregated data are used instead of individual-level data as the treatment varies at this level. The aggregate data are preferred in order to avoid problems of within-region correlations in the error term that may otherwise result in underestimated standard errors as Donald and Lang (2007) show. Using raw aggregated data, as is done in this case, yields results similar to those when using the residual aggregation method and hence adjusting for background characteristics available in the data as suggested by, e.g., Bertrand, Duflo, and Mullainathan (2004).

[29] See Doblhammer and Vaupel (2001) and Buckles and Hungerman (2008) on the importance of controlling for season of birth.

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TABLE 1
Estimated Prenatal Exposure to the Policy Experiment

Exposure Group	Month of Birth	Date of Conception	Gestational Age at Start of Experiment (Months)	Number of Weeks In Utero during Experiment		Trimester Underexposure	Experiment May Have Affected Conception Rate?
				Minimum	Maximum		
1	Before Nov. 1967	Before Feb. 1967	Born	0	0	. . .	No
2	Nov. 1967	Feb. 1967	8–9	0	4	3	No
2	Dec. 1967	March 1967	7–8	4	8	3	No
2	Jan. 1968	April 1967	6–7	8	12	3	No
2	Feb. 1968	May 1967	5–6	12	16	2, 3	No
2	March 1968	June 1967	4–5	16	20	2, 3	No
3[a]	April 1968	July 1967	3–4	20	24	2, 3	No
3	May 1968	Aug. 1967	2–3	24	28	1, 2, 3	No
3	June 1968	Sep. 1967	1–2	28	32	1, 2, 3	No
3	July 1968	Oct. 1967	0–1	32	34	1, 2, 3	No
4	Aug. 1968	Nov. 1967	NC	30	34	1, 2, 3	Yes
4	Sept. 1968	Dec. 1967	NC	26	30	1, 2, 3	Yes
4	Oct. 1968	Jan. 1968	NC	22	26	1, 2, 3	Yes
4	Nov. 1968	Feb. 1968	NC	18	22	1, 2	Yes
4	Dec. 1968	March 1968	NC	14	18	1, 2	Yes
4	Jan. 1969	April 1968	NC	10	14	1, 2	Yes
4	Feb. 1969	May 1968	NC	6	10	1	Yes
4	March 1969	June 1968	NC	2	6	1	Yes
4	April 1969	July 1968	NC	0	2	1	Yes
5	After April 1969	After July 1968	NC	0	0	. . .	No

Note.—The experiment started on November 1, 1967, and ended on July 14, 1968. The date and gestational age at the start of the experiment all assume that conception occurred 9 months prior to birth. NC = not conceived.

[a] Exposure group 3 is the treatment group in the main analysis.

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composition of young mothers; $\eta_{t,m<21}$ are period by mother age at birth effects, which besides general time effects also account for potential changes in the impact of having a young mother across the cohorts. For example, during the observation period the number of mothers under the age of 21 decreased, and hence, the composition of these mothers may have changed in terms of, for example, parental skills. The period by mother age effects account for such and similar compositional changes. The expression $\varepsilon_{r,t,m<21}$ is the error term. All regressions are weighted by the number of children in each cell. The reported standard errors are robust with respect to heteroscedasticity.

I also provide estimates from various versions of equation (3). In particular, when considering the impact on early health and selection into the adult outcome sample, I report estimates from

$$Y_{r,t,m<21} = \alpha_0 + \sum_{i=1}^{5} \beta_i \text{EXPOSURE}_{i,r,t,m<21} + \eta_{r,t} + \eta_{r,m<21} + \eta_{t,m<21} + \varepsilon_{r,t,m<21}, \tag{5}$$

which captures the effects on exposure groups 1–2 and 4–5 as well.

The DDD model accounts for many possible confounders and perhaps, most importantly, region-specific shocks coinciding with the experiment also affecting the children's outcomes. Hence, in order for a contemporary local shock to bias the estimate of β_3, not only must the timing of the temporary unobserved shock precisely coincide with the timing of the temporary policy experiment, but it must also affect the adult labor market outcomes only of children born to mothers under the age of 21 and not children born to older mothers. While it is impossible to directly test this assumption, in the following sections, besides the baseline DDD estimates, results from a number of robustness checks assessing the plausibility of this identifying assumption are also reported. Moreover, there are no indications that a shock (or other policy changes) fulfilling these conditions occurred at the same time as the policy experiment.[30]

In addition, note that the use of month of birth data and the fact that children born during the same calendar year typically start school at the same time implies that the estimated impact of the policy will not likely be biased by peer effects. Disruptive behavior of a few exposed classmates will not bias the estimate unless the peers affect classmates born only in the same period and not earlier or later in the same year. Moreover, in

[30] Note also that the same conditions must hold in order for a common shock later in life to bias the estimates. This is important since most other shocks (say major plant closures) that conceivably could affect adult outcomes are likely to affect cohorts born just before, during, or after the experiment in a smooth manner with respect to birth cohort.

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the analysis I also consider outcomes that are not likely to be strongly affected by peers, such as fluid intelligence test scores.

If data on the exact differences in changes in alcohol consumption across pregnant mothers of the two age groups were available, the estimated β_3 parameter in equation (4) could be scaled to reflect, for example, the impact per unit of alcohol. Accurate data on alcohol consumption during pregnancy are, however, difficult to attain because of recall or desirability biases.[31] Without the exact magnitudes in alcohol consumption change, the β_3 estimate reflects the impact of alcohol availability on the children born to mothers under age 21 in the experiment regions. Naturally, the relevance of this parameter from an alcohol policy perspective hinges on the response of young pregnant mothers to a similar change in availability today. The policy experiment took place well before the widespread information campaigns about the potential detrimental effects of alcohol exposure on fetal development.[32] Hence, it seems likely that such information may reduce the impact of a similar change in alcohol availability today. On the other hand, the lack of information may also imply that the estimated impact of the policy provides an estimate closer to an average population effect of an increased prenatal alcohol exposure than what a similar policy experiment may have been able to provide today.

In any case, since it is impossible to rule out alternative biological explanations associated with a temporary increase in alcohol availability besides alcohol exposure (e.g., a temporary increase in smoking), the estimated effects should be interpreted as the impact of the increase in alcohol availability, and only carefully as the impact of prenatal alcohol exposure per se. Similar alternative biological explanations are endemic to both the literature on prenatal alcohol exposure in humans and the literature on the developmental origins of economic and health outcomes later in life. Despite these limitations, the current analysis significantly adds to both these literatures.

V. Results

This section starts by providing descriptive statistics for the main sample. I then present the results for the immediate impact on child health and

[31] Studies of alcohol-related birth defects in humans rely heavily on maternal self-reports of alcohol use. Several studies have demonstrated that drinking during pregnancy is often underreported (Ernhart et al. 1988; Alvik et al. 2006). Clark, Dawson, and Martin (1999) found that depending on the screening tool used, alcohol use among pregnant women ranged from 21 percent to 70 percent.

[32] Online app. B provides a brief scientific history of studies on prenatal alcohol exposure. See also App. fig. A1 and online fig. B1 for further evidence on the diffusion of information in the science communities and among the public.

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for selection into the adult outcome sample based on family background characteristics. Section C provides the main results. The following sections provide additional evidence on cognitive and noncognitive skills and robustness checks.

A. *Descriptive Statistics*

The first panel of table 2 reports the mean labor and educational outcomes for children born in the treatment and control regions. Columns 1–6 report averages for children born in the experimental regions (cols. 1–3) and the control regions (cols. 4–6) to mothers above age 21. Columns 7–12 report the corresponding characteristics for children of mothers under age 21 at delivery.[33] The statistics in table 2 are calculated for the cohorts born during the first two quarters of each year and hence do not differentiate between cohorts depending on their duration or timing of exposure. It seems that the children of the young mothers exposed to the experiment (i.e., born in 1968) tend to have less favorable development in terms of educational and labor market outcomes compared to the other cohorts. The second panel presents descriptive statistics for grandpaternal income in 1968, grandmaternal family size, and the average number of children in each cell. A slight increasing age trend among young mothers and decreasing trend in the number of young mothers can be noted in both the treatment and the control regions.

B. *The Impact on Early-Life Health and Selection into the Adult Sample*

Before I turn to the main results, table 3 examines the immediate impact of the policy on prenatal health. Column 1 shows that, consistent with a higher male vulnerability to early shocks, the share of males is 7.3 percentage points lower among the children exposed to the policy from the first half of the pregnancy. Columns 2 and 3 show that this is driven by a reduction in the male cohort size. Column 4 shows that the impact on share of males is concentrated among those exposed from the first trimester. In the absence of data on gestational age at birth, columns 5 and 6 present estimates when using as the dependent variable the average calendar month of birth for children born between January and July (i.e., months 1–7) in each year. Exposed males were, on average, born about 1 week earlier (0.28 month), but there is no change in the month of birth of females. Appendix table A2 provides the full set of equation (5) estimates

[33] All averages are calculated using data aggregated to the region by month of birth by old/young mother level and weighted by the number of children in each cell.

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TABLE 2
TRENDS IN BACKGROUND CHARACTERISTICS AND OUTCOMES OVER THE OBSERVATION PERIOD

	Mother Age ≥ 21						Mother Age < 21					
	Treated Regions			Control Regions			Treated Regions			Control Regions		
	Born <1968 (1)	Born 1968 (2)	Born >1968 (3)	Born <1968 (4)	Born 1968 (5)	Born >1968 (6)	Born <1968 (7)	Born 1968 (8)	Born >1968 (9)	Born <1968 (10)	Born 1968 (11)	Born >1968 (12)
	A. Child Outcomes											
Yearly earnings at age 32 (in 100 SEK)	7.194	7.356	7.393	7.205	7.329	7.411	7.014	7.113	7.258	7.055	7.231	7.252
Share with zero earnings (age 32)	.105	.075	.092	.098	.089	.085	.137	.143	.129	.129	.099	.115
Share on welfare in 2000	.028	.021	.036	.025	.023	.027	.068	.078	.089	.056	.057	.071
Education (years)	12.71	12.88	12.98	12.69	12.79	12.99	11.67	11.62	11.81	11.71	11.82	11.76
Share high school graduates	.916	.938	.924	.918	.932	.932	.824	.807	.831	.824	.845	.823
Share college graduates (≥ 2 years)	.332	.364	.402	.342	.361	.410	.161	.163	.191	.184	.199	.190
Share of males	.51	.52	.51	.51	.51	.52	.50	.49	.53	.51	.52	.52
	B. Background Characteristics											
Grandfather income in 1968 (in SEK)	10.02	10.04	10.12	10.01	10.05	10.11	9.974	10.02	10.02	9.972	10.00	9.991
Completed grandmaternal fertility	2.158	2.099	2.116	2.125	2.114	2.117	2.542	2.507	2.537	2.523	2.521	2.576
Age of mother at delivery	25.65	25.56	25.58	25.45	25.42	25.58	18.89	18.92	18.93	18.89	18.95	18.93
Age of father at delivery	28.65	28.29	28.07	28.48	28.15	28.14	22.54	22.40	22.98	22.53	22.45	22.80
Average number of children per quarter cell	664	665	686	447	447	460	232	194	153	159	132	103

NOTE.—The table reports cell averages for children born during the first two quarters of each year for the treatment and control regions. All statistics are weighted by the number of children in each cell, except for the cohort sizes, which report the unweighted mean.

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TABLE 3
THE IMPACT ON EARLY-LIFE HEALTH AND SELECTION INTO ADULT OUTCOME SAMPLE

	Full Sample: All						1968 Family Income Sample				
							All			High-Income Family	Low-Income Family
Policy Exposure Group	Share of Males (1)	Male Cohort Size (2)	Female Cohort Size (3)	Share of Males (4)	Month of Birth Males (5)	Month of Birth Females (6)	Grandfather Income 1968 (7)	Share of Males (8)	Share of Males (9)	Share of Males (10)	Share of Males (11)
Exposed from 1st half of pregnancy	−.073**	−.176*	.093				.003	−.076**		−.059	−.090**
	(.0309)	(.1158)	(.078)				(.032)	(.035)		(.044)	
Exposed from 3rd trimester				.008					.007		
				(.041)					(.036)		
Exposed from 2nd trimester				−.029					−.026		
				(.029)					(.033)		
Exposed from 1st trimester				−.082***					−.081**		
				(.026)					(.034)		
Exposed in utero (groups 2 and 3)					−.28*	.056					
					(.134)	(.141)					
Observations	4,104	4,104	4,104	4,104	342	342	4,104	4,104	4,104	4,104	4,104
Mean of dependent variable (monthly cells)	.514	43.96	41.47	.514	4.05	4.05	9.98	.513	.513	.514	.513

NOTE.—Each column represents a separate estimate from eq. (4). Log cohort size coefficients are adjusted using $\exp(\beta) - 1$. See App. tables A2 and A3 for the full set of estimates. Columns 1–3 and 7–9 report estimates from the model in eq. (4). Columns 4 and 10 report extended versions of eq. (4) allowing the effects of the policy to vary by estimated gestational trimester at the start of the policy experiment. Columns 5 and 6 report estimates from a specification with the month of birth as the dependent variable only on children born before August each year and replace the year × month effects with year effects.

* Significant at 10 percent.

** Significant at 5 percent.

*** Significant at 1 percent.

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showing that there are no significant effects on the other exposure groups. Table A2 shows that the results are similar when restricting the sample to children of young mothers.

The identification strategy seeks to avoid bias from selection into pregnancy. However, given the effects on early-life health, it is of interest to understand if exposure influenced selection into the surviving adult sample. This can be assessed using predetermined family characteristics, which are difficult to find for the young parents during this period. However, for 78 percent of the sample, it is possible to proxy for family economic status using grandfather's income in the 1968 Income and Taxation register.[34] Table 3, column 7, shows that there is no correlation between exposure from early pregnancy and grandfather's income. Columns 8 and 9 show that the impact on the share of males holds in the grandfather's income sample. Columns 10 and 11 show that the effects on share of males are similar in high- and low-income families.

Appendix tables A3 and A4 provide the estimates for children born just before or after or conceived during the policy. Consistent with indirect effects on family composition, children conceived early during the policy have grandfathers with an average 8 percent lower income.[35] Moreover, children conceived early during the policy are 16 percent more likely to have a younger half sibling. When interpreted through the lens of the marital instability model of Becker, Landes, and Michael (1977) or Weiss and Willis (1997), the latter result suggests that children conceived during the policy were born into less stable families, since at least one parent was more likely to subsequently have a child with a different partner.[36] Note that the timing of the family composition effects coincides with the timing of the increase in drunkenness arrests and the effects on health-related outcomes among those in early gestation at the start of the policy.

[34] Data are available for those who have a living grandfather under age 67 (median age 55) in 1968 and are linked to the exposed children using the multigenerational register. Children are included if either the maternal (67 percent) or paternal (27 percent) grandfather is matched, replacing the maternal fathers' income with the paternal fathers' income if the former was missing.

[35] Grandparents' educational attainments are also available for those whose grandparents were born after 1925 and were alive in 1990 (i.e., 65 and under in 1990). However, for this outcome it is necessary to restrict the sample to the children born to young mothers. In practice it limits the sample to the under age 21 women who themselves were born to women under age 20 at birth. The estimates for this limited sample are similar to the grandfather's income sample results and are also presented in App. table A3.

[36] In these models poorer initial match quality and increased accidental conceptions, respectively, increase family dissolutions. The mechanism I have in mind is increased alcohol consumption ⇒ reduced assortative mating (increased random matching) ⇒ poorer initial match quality ⇒ higher risk of separation after childbearing ⇒ higher probability of subsequent multiple partner fertility. Multiple partner fertility is also strongly negatively correlated with children's outcomes, and many sociological studies have found that it is correlated with lower paternal investments in children.

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In summary, the policy seems to have influenced the prenatal health of the children exposed from early pregnancy, particularly among males.[37] Moreover, the increased alcohol availability affected the composition of children conceived early during the policy.[38] However, I find no indications of sample selection effects among those conceived before the policy started, which is important for the interpretation of the remainder of the results.

C. Main Results

1. Graphical Difference-in-Differences

Figure 2 plots the average log earnings at age 32 for the children born around the time of the policy, split by maternal age at birth.[39] Relative to the control regions, there is a distinct drop in earnings in the treatment regions that coincides with the timing of the policy among children born to young mothers (panel A), while no such pattern is seen for the older mothers (panel B). The same pattern is evident for years of schooling (App. fig. A2).

To get a first clue about where in the distribution the change in average earnings stems from, figure 3 plots the differences in earnings distributions for the most exposed cohort. The left-hand side of panel A shows the cumulative earnings distribution (cdf) of women and men born to young mothers during the second quarter of 1968. The cdfs suggest that the lower end of the distribution seems to have been particularly strongly affected as the distribution is pushed to the left for the exposed cohort, particularly among the males. In contrast, the earnings differences between those born in the control and treatment regions earning above the 50th percentile are smaller. Under the assumption that in the absence of the policy the treated children would have ended up at the same position of the distribution, the policy affected low–socioeconomic status (SES) males in particular.

The right-hand side of figure 3 shows that for those born 1 year before the policy, the difference in distribution between the control and treat-

[37] This is consistent with the correlations between prenatal alcohol exposure and birth outcomes found by Little et al. (1986, 277) indicating "a greater vulnerability of the male to alcohol exposure in the late first and early second trimester," as measured by birth weight. Interestingly, although not noted by the authors, the fraction of male births in their sample is also strongly negatively correlated with consumption during the same period of gestation. Furthermore, the results are consistent with differences in sensitivity to binge alcohol exposure in animals (Goodlett and Peterson 1995).

[38] Compositional effect may also explain the absence of sex ratio effects for those conceived early during the policy.

[39] The data used in the figure are expressed in year 2000 Swedish kronor (SEK). Individuals with earnings below the 1st percentile (SEK 1,338) and above the 99th percentile (SEK 538,004) are omitted.

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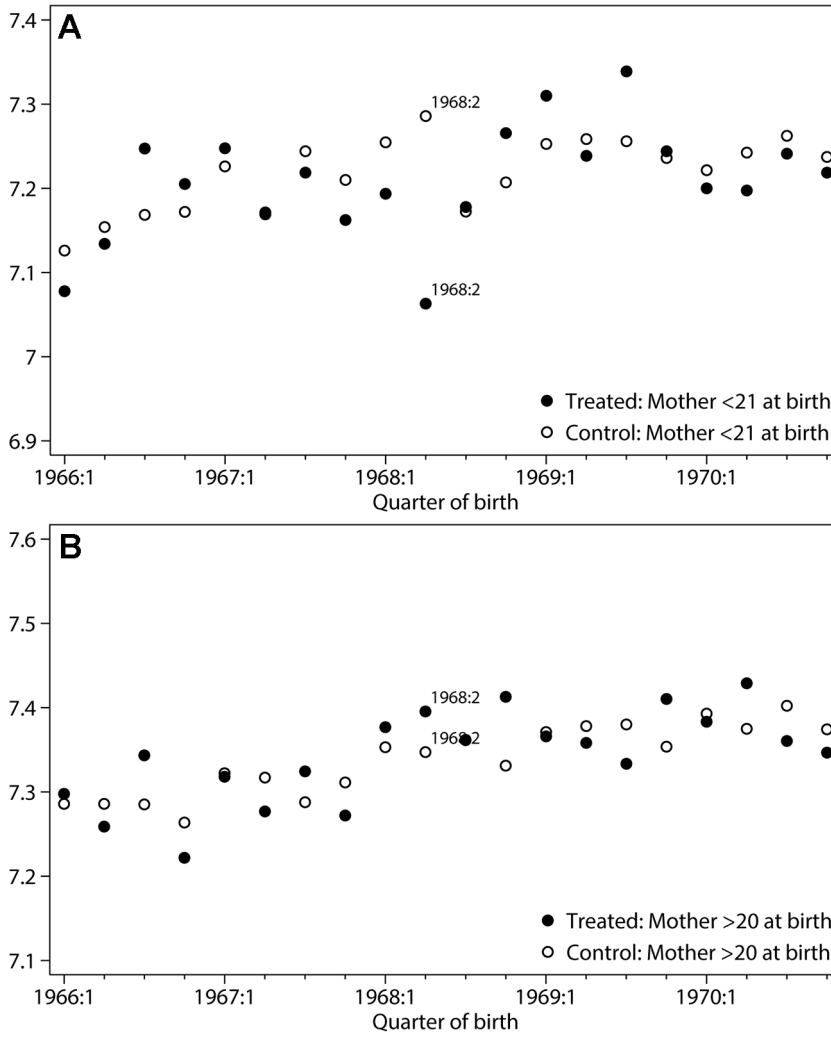


FIG. 2.—Average (log) earnings at age 32 (all children), treated versus control, by age of mother at birth. Average earnings by quarter of birth for children born in treated and control regions to mothers under age 21 (panel A) and for children born to mothers older than 21 at birth (panel B).

ment regions is minimal. This is also the case for children of the mothers above age 21 (panel B).

2. Baseline Results

Table 4 reports the estimates of β_3 in equation (3) using average log earnings, the share with no labor income, and the share receiving welfare as

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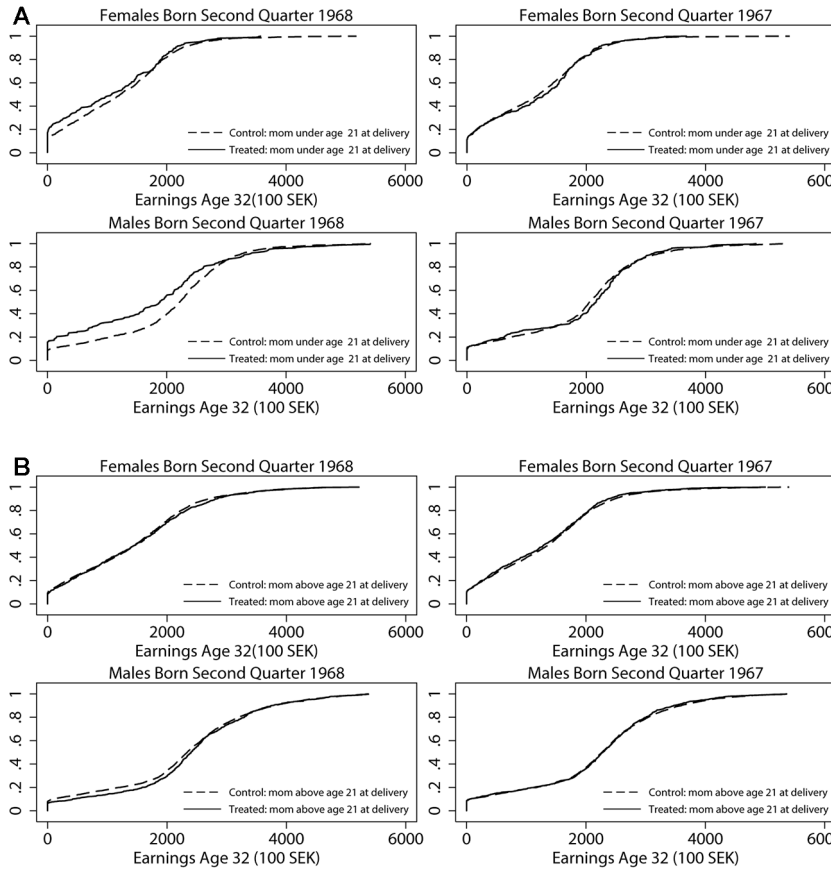


FIG. 3.—Cumulative earnings distribution at age 32. The left column presents earnings for women (top) and men (bottom) born during the second quarter of 1968. The right column shows the same distributions for children born during the second quarter of 1967 (i.e., before the experiment). Panels A and B show separate distributions by maternal age at birth.

the dependent variables. Columns 1–3 provide the estimates for the full, male, and female samples, respectively. Panel A shows that, on average, the exposed cohort has 24 percent lower earnings, with larger and more precise estimates for males than for females.[40] Panel B shows a significant increase in the share with no labor income (8.3 percentage points) for men and an insignificant (6 percentage points) increase for women.

[40] However, women's earnings at the age of 32 may not accurately reflect their permanent earnings. Böhlmark and Lindqvist's (2006) estimates of life cycle biases show that, in the case of Sweden, the ideal solution for women would be to use earnings after the age of 40 in order to get a good proxy for permanent earnings. Indeed, if using the age 37 earnings or the total earnings between ages 26 and 37, the effects are similar for males but smaller for females.

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TABLE 4
IMPACT ON LABOR MARKET OUTCOMES

	Subsample		
	All (1)	Men (2)	Women (3)
	A. Dependent Variable: Average Log Earnings		
Exposed from 1st half of pregnancy	−.244***	−.241**	−.174
	(.090)	(.113)	(.138)
Adjusted R^2	.50	.45	.27
Mean of dependent variable	7.133	7.440	6.801
	B. Dependent Variable: Share with Zero Earnings at Age 32		
Exposed from 1st half of pregnancy	.072***	.083***	.059
	(.027)	(.024)	(.043)
Adjusted R^2	.23	.15	.08
Mean of dependent variable	.123	.112	.135
	C. Dependent Variable: Share on Welfare in 2000		
Exposed from 1st half of pregnancy	.035***	.045**	.027**
	(.012)	(.022)	(.012)
Adjusted R^2	.26	.07	.10
Mean of dependent variable	.0624	.0569	.0682
Year by month of birth dummies	Yes	Yes	Yes
Region of birth dummies	Yes	Yes	Yes
Mother under age 21 dummy	Yes	Yes	Yes

NOTE.—Each column and panel represents a separate regression; $N = 4{,}104$. The unit of observation is all firstborn children alive in 2000 to mothers either age ≥ 21 or below in a given year, quarter, and region. Exposed from 1st half of pregnancy is a dummy equal to one if the child was born to a mother under age 21 and exposed to the experiment while in utero from early until late pregnancy (see Sec. IV for details). All regressions include year by month of birth, region of birth, and mother under age 21 at delivery dummies and a set of interaction terms between these variables (see eq. [4]). All regressions are weighted by the cell size used to calculate the dependent variable. The earnings coefficients presented are given by the transformation $\exp(\beta) - 1$). Heteroscedasticity robust standard errors are reported in parentheses.

* Significant at 10 percent.
** Significant at 5 percent.
*** Significant at 1 percent.

Panel C reveals that the proportion receiving welfare is 4.4 percentage points higher for men and 2.9 percentage points for women. Relative to the mean, the impact on welfare is almost twice as large for males as for females. Panel A of Appendix table A5 presents the estimates for the other exposure groups as well. As for the health-related outcomes, the effects are concentrated among children in early pregnancy at the start of the policy. Panel B of table A5 shows that this conclusion does not change when restricting the sample to children born to young mothers (i.e., difference-in-difference).

In summary, the estimates confirm the pattern in the figures. The effects on labor market outcomes among those conceived before but ex-

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posed to the policy from early pregnancy are substantial, with larger and more precise estimates for males than for females.

3. Distributional Impact

Figure 4 provides unconditional quantile estimates (Firpo, Fortin, and Lemieux 2009) to provide a better understanding of the roots of the mean impact on earnings. Consistent with the pattern in figure 3, figure 4 shows that the mean effect on earnings to a large extent is driven by changes in earnings below the 40th percentile of the earnings distribution. Above the 40th percentile, the point estimates hover around −14 percent up to the 80th percentile, after which the effect goes toward zero. This suggests that much of the mean impact on earnings stems from a reduction in the number of hours worked/employment.

Consistent with this interpretation, the estimated impact on monthly full-time wages (available for 37 percent of the sample) shows a distinctly different pattern.[41] The mean impact on wages is −3.5 percent; however, as shown in figure 4, this stems from a large decrease (~−9 percent) in the middle of the distribution and smaller effects toward the tails. The absence of wage effects in the lower tail of the distribution is likely partly due to the positive selection into the wage sample (as suggested by the earnings figure), but the wage-setting institutions, with industry-specific minimum wages, likely also play a role. In the lower tail of the distribution, wages are typically collectively bargained, while individual wage bargaining is more common higher up in the distribution (National Mediation Office 2011). Again, as for earnings, in the top of the distribution there is not much evidence of an impact of the policy.

Finally, the welfare state setting seems important to keep in mind for understanding the impact on labor earnings. This is illustrated by the comparatively muted, relative to earnings, impact on disposable income also shown in figure 4.

In summary, negative earnings effects are found throughout the distribution. The earnings effects likely reflect a reduction in the annual hours worked in the lower half of the distribution. Sizable reductions in earnings are, however, also found higher up in the distribution where lower wages play a more important role. A benign interpretation of the differences in the effects on earnings and disposable income is that the welfare state to a large extent has compensated for the policy's effect on productivity. On the other hand, it is also possible that the generous welfare state

[41] The wage data are collected in *Strukturlönestatistiken* by Statistics Sweden and cover all employees in the public sector, but they are sampled at the firm level for the private sector. The sampling probability depends on firm size, implying that small firms are underrepresented. Conditional on being employed in the month of November, all individuals in the sampled firms are covered. The quantile estimates for earnings are very similar in the wage sample.

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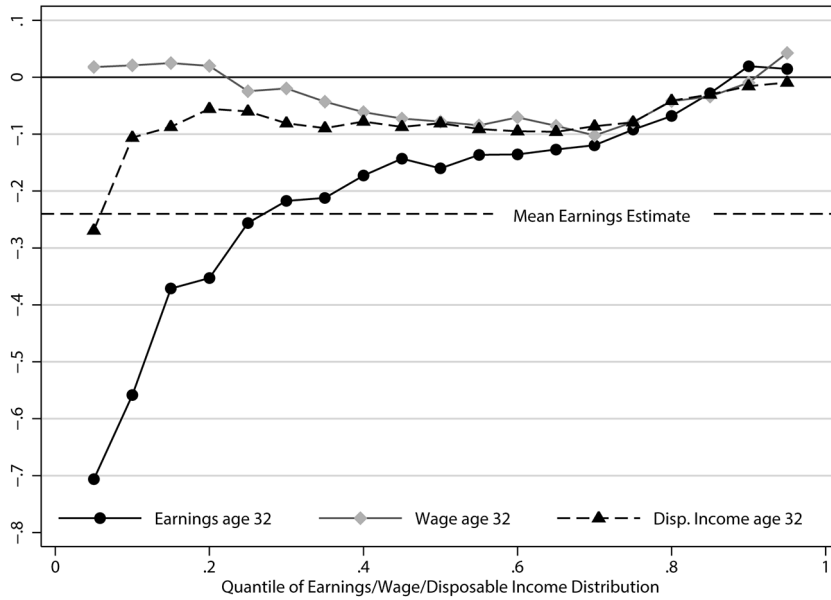


FIG. 4.—Earnings, wages, and disposable income distribution effects. Monthly wages for males and females are expressed in full-time equivalents during the sampling month (September, October, or November depending on the sector of employment). The wage data are based on complete data from all employees in private firms with more than 500 employees and all public-sector employers. Smaller private firms are sampled randomly, and all employees in the sampled firms are included. In total, age 32 wage data are available for 37 percent of the full sample. Earnings and disposable income are available for the full sample and measure total annual income. The figure reports unconditional quantile regression estimates (Firpo et al. 2009) for these outcomes using the individual-level data version of the specification in equation (4).

arrangements may have exacerbated the effects on earnings by weakening labor supply incentives, by lowering demand for low-skilled workers through high minimum wages, or by weakening parents' investment incentives (Lindbeck and Nyberg 2006). It is, however, difficult to quantify the role of these mechanisms with the data at hand. The following sections provide insights into the role of other mediators for which data are readily available.

D. Mediators

1. Education

Similar to the effects on earnings, the impact on education is substantial. Panels A, B, and C of table 5 report estimates for average years of schooling, the proportion of high school graduates, and the proportion with at least 3 years of higher education as dependent variables. Columns 1–3 in each panel provide the estimates for β_3 in the full sample, the male sam-

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TABLE 5
IMPACT ON EDUCATION

	Subsample		
	All (1)	Men (2)	Women (3)
	A. Dependent Variable: Years of Schooling		
Exposed from 1st half of pregnancy	−.309***	−.516***	−.209*
	(.089)	(.167)	(.121)
Adjusted R^2	.85	.78	.73
Mean of dependent variable	11.78	11.52	12.05
	B. Dependent Variable: Share of High School Graduates		
Exposed from 1st half of pregnancy	−.063***	−.098***	−.033
	(.016)	(.023)	(.025)
Adjusted R^2	.57	.40	.36
Mean of dependent variable	.834	.821	.849
	C. Dependent Variable: Share Graduated from Higher Education		
Exposed from 1st half of pregnancy	−.032*	−.062**	−.030
	(.018)	(.028)	(.023)
Adjusted R^2	.79	.69	.65
Mean of dependent variable	.193	.138	.251
Year by month of birth dummies	Yes	Yes	Yes
Region of birth dummies	Yes	Yes	Yes
Mother under age 21 dummy	Yes	Yes	Yes

NOTE.—Each column and panel represents a separate regression; $N = 4{,}104$. The unit of observation is all firstborn children alive in 2000 to mothers either aged ≥ 21 or below in a given year, quarter, and region. Exposed from 1st half of pregnancy is a dummy equal to one if the child was born to a mother under age 21 and exposed to the experiment while in utero from early until late pregnancy (see Sec. IV for details). All regressions include year by month of birth, region of birth, and mother under age 21 at delivery dummies and the appropriate interaction terms between these variables (see eq. [4]). All regressions are weighted by the cell size used to calculate the dependent variable. Heteroscedasticity robust standard errors are reported in parentheses.

* Significant at 10 percent.

** Significant at 5 percent.

*** Significant at 1 percent.

ple, and finally the female sample. In the full sample, policy exposure significantly reduced years of schooling by, on average, −0.32 year, −0.52 among males, and −0.22 for females (but then imprecisely estimated). On average, high school completion rates are 6.7 percentage points lower, which is driven by a 13 percent lower high school completion rate among males (−0.105/0.82). The share graduating from higher education is reduced by 5.9 percentage points among men and by 3.1 percentage points among women (imprecisely estimated). Unconditional quantile estimates (App. fig. A3) suggest that the reduction in educational attainment primarily stems from a substantial decrease in the likelihood of completing high school and attending college.

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2. Cognitive and Noncognitive Ability

For males it is possible to assess cognitive and noncognitive ability directly using military enlistment data. Conscription was mandatory for males, and almost all men (90 percent) who were not in too poor health were enlisted in the military service. Enlistment took place at age 18 or 19. Over 2 days the enlistees go through tests to measure their medical and physical status and their cognitive ability and an interview with a psychologist.[42]

The general cognitive ability (*G*) test score is a standardized 9-point scale, and it is based on four subtests designed to capture fluid (*Gf*; logic and spatial tests) and crystallized (*Gc*) intelligence (synonyms and technical comprehension tests).[43] The noncognitive score is based on an approximately 20–25-minute interview with a psychologist resulting in four different scales, all ranging from 1 to 5: (i) social maturity (extraversion, having friends, taking responsibility, independence), (ii) psychological energy (perseverance, ability to fulfill plans, ability to remain focused), (iii) intensity (the capacity to activate oneself without external pressure, the intensity and frequency of free-time activities), and (iv) emotional stability (ability to control and channel nervousness, tolerance of stress, and disposition to anxiety). The noncognitive subscore tests are combined into a standardized 9-point scale that has a discrete approximation to a normal distribution (see Bihagen, Nermo, and Stern 2013).

I use the scores in three ways, as a standardized continuous measure (mean zero, standard deviation one) and as three variables capturing the share of the sample with low, medium, and high scores. As can be seen in table 6, prenatal exposure to the policy decreases both average cognitive and noncognitive test scores, and again the impact is concentrated in the lower tail of the distribution. The share with a cognitive ability score in the bottom third increases by 27 percent and by 16 percent for the noncognitive ability. Noncognitive skills have previously been found to be particularly important at the lower end of the earnings distribution (Heckman, Stixrud, and Urzua 2006; Lindqvist and Vestman 2011).[44]

Appendix table A6 provides the effects on the four cognitive and noncognitive subscores separately. While the *Gf* tests are similarly influenced,

[42] See Lindqvist and Vestman (2011) for a detailed account of the enlistment procedure for the cohorts born between 1965 and 1974 and how the cognitive and noncognitive test scores relate to labor market outcomes.

[43] The *Gf* tests reflect the capacity to think logically and solve problems in novel situations independent of acquired knowledge. The *Gc* tests focus on verbal ability and making use of acquired knowledge (Cattell 1971, 1987). See Carlsson et al. (2013, fig. A1) for examples of the subsample questions.

[44] In a joint estimation on the full sample, the five strongest predictors of zero income are low social maturity (+48 percent), low psychological energy (+42 percent), low emotional stability (+41 percent), low logic (+24 percent), and low intensity (+16 percent).

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TABLE 6
EFFECTS ON COGNITIVE AND NONCOGNITIVE ABILITY

	Average Score (1)	*P*(Low Score) (2)	*P*(Medium Score) (3)	*P*(High Score) (4)
	A. Cognitive Ability			
Exposed from 1st half of pregnancy	−.053	.069***	−.052***	−.002
	(.070)	(.025)	(.020)	(.029)
R^2	.88	.69	.29	.85
Mean of dependent variable (young mothers)	−.224	.257	.585	.167
	B. Noncognitive Ability			
Exposed from 1st half of pregnancy	−.099	.030**	−.013	−.016
	(.096)	(.013)	(.050)	(.066)
Adjusted R^2	.74	.56	.42	.68
Mean of dependent variable (young mothers)	−.113	.188	.640	.207

NOTE.—The table reports eq. (4) estimated effects on cognitive and noncognitive skills score as graded by trained psychologists at military enlistment procedure at age 18 for males. For ease off comparison, all scores have been standardized, with mean zero and standard deviation one. The share low, medium, and high scores are defined on the basis of the original scores, which takes on integer values between 1 and 5, with 5 being the highest score for the noncognitive score and 1–9 for the cognitive ability score. For noncognitive ability, the respective category is defined as low (1–2), medium (3), and high (4–5). For cognitive ability, the score is defined as low (1–3), medium (4–6), and high (7–9).

* Significant at 10 percent.

** Significant at 5 percent.

*** Significant at 1 percent.

the impacts on the *Gc* tests are mixed. The synonyms test score is negatively affected similarly to (if not even more than) the logic and spatial tests, while the technical comprehension test even shows signs of improvement.[45] The four noncognitive subscores show some heterogeneity as well. The negative effects on "social maturity" and "intensity" are comparable to the effect on the fluid intelligence test scores, but "psychological energy" and "emotional stability" are not significantly affected.[46]

[45] School track selection based on absolute or comparative advantages could potentially explain these results. Crystallized intelligence is more malleable than fluid intelligence. As a result of the poorer performance in school, the exposed children may have opted for more vocational-oriented tracks rather than academic tracks. This could have further reduced their word comprehension but potentially increased their technical comprehension skills.

[46] The noncognitive subscores have previously been used only in the context of men in elite positions (Bihagen et al. 2013) and by Fredriksson, Hensvik, and Nordström-Skans (2015), who show that the subscores are valued independently in the labor market. See online app. table D1 for a classification of the subscores in terms of the Big Five traits of personality.

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3. Decomposing the Effects of the Policy on Outcomes by Source

Overall, the results for the mediating factors correspond well with the effects on earnings with the largest effect found in the lower tail of the distribution. Using a decomposition method similar in spirit to the method in Heckman, Pinto, and Savelyev (2013), described in detail in online appendix C, figure 5 show the relative contribution of the cognitive and noncognitive skills and other (unmeasured) factors to the total effect (normalized to 100 percent).

The decomposition suggests that noncognitive skills play a larger role for the labor market outcomes and cognitive skills for educational outcomes. The policy's impact on the average measured skills can explain about 13 percent of the policy's impact on labor market outcomes and 18–30 percent of the impact on educational outcomes. For comparison, Heckman et al. (2013) find that for males the impact of the Perry Preschool Program on "externalizing behavior" accounts for about 19 percent of the total treatment effect on income and the probability of em-

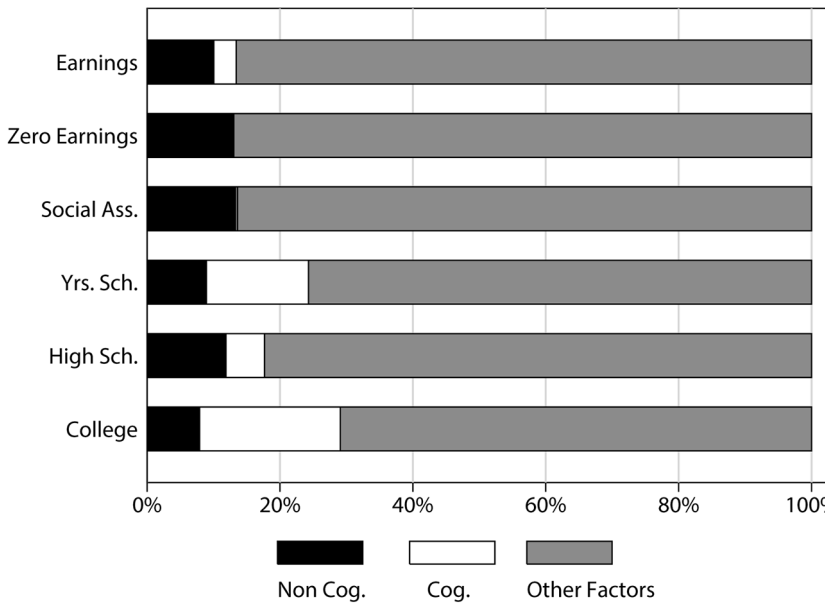


FIG. 5.—Decomposition of the effect of the policy on outcomes for males. The figure presents the estimated decompositions of the impact of the policy into policy-induced reductions in cognitive, noncognitive, and other factors. By "other factors" I mean the residual effect associated with unmeasured skills. See online appendix C for detailed information on the procedure to produce the figure. Cognitive and noncognitive skill data are not available for females, so estimates show the decomposition for enlisted males only.

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ployment and that "academic motivation" accounts for about 18 percent of the impact on the California Achievement Test.[47]

Obviously, even with these relatively detailed skill measures, it is hard to fully account for the impact of the policy. There are, however, several reasons why this simple decomposition should be interpreted with care. First, much of the action in cognitive and noncognitive skills takes place in the lower tail of the distribution (as shown in table 6); hence the mean changes in cognitive and noncognitive skills used to construct figure 5 may not be all that informative. Second, the decomposition assumes that the measured skills are independent of each other and the unmeasured skills (other factors). Third, although I use the reliability ratios estimated by Grönqvist, Öckert, and Vlachos (2010) to rescale the estimated effects of cognitive and noncognitive skills, there is still a lingering concern of measurement error bias. A better alternative would be to follow Heckman et al. (2013), who address both the endogeneity and measurement error issues.

E. Heterogeneity and Robustness Checks

1. Estimates by Family Income

Table 7 presents the results after splitting the data by grandfather's 1968 income. The effects are consistently more negative for children from low-income families than from high-income families, suggesting that higher parental resources cushion the effects of negative health shocks (Currie and Hyson 1999; Currie 2009). It is, however, difficult to rule out that the effects across SES background at least in part are due to differences in consumption responses. However, to the extent that SES differences in prenatal conditions are reflected in the impact on the sex ratio, Appendix table A3 suggested that prenatal conditions were similarly affected across income and education groups.[48]

2. The Timing of Exposure

In order to attain an even clearer picture of the timing of the impact of the policy, I let the 4-month treatment window glide over the cohorts born between January 1964 and December 1972. Figure 6 plots the locally weighted average of the standardized estimated triple-difference parameter (β_3) for education and earnings. The estimates between the two ver-

[47] The California Achievement Test measures reading, spelling, language, and mathematics skills.

[48] It is also possible that the same shock will have a greater impact among children in low-SES families if they are on a steeper portion of the production function (Almond and Currie 2010).

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TABLE 7
LONG-TERM EFFECTS BY FAMILY INCOME

	Outcome Variables					
	(ln) Earnings (1)	Zero Earnings (2)	Welfare (3)	Years of Schooling (4)	High School (5)	University (6)
	A. Baseline Estimates for the Family Income Sample					
Exposed from 1st half of pregnancy	−.269***	.074***	.041***	−.464***	−.081***	.057***
	(.105)	(.024)	(.012)	(.111)	(.020)	(.018)
Mean of dependent variable	7.14	.11	.06	11.79	.837	.194
	B. Estimates by Grandfather's 1968 Income					
High-income family:						
Exposed from 1st half of pregnancy	−.142*	.038	.038*	−.327*	−.060	−.046*
	(.084)	(.033)	(.022)	(.167)	(.039)	(.028)
Mean of dependent variable	7.167	.112	.056	11.89	.849	.209
Low-income family:						
Exposed from 1st half of pregnancy	−.38***	.110***	.044**	−.614***	−.107***	−.057***
	(.149)	(.036)	(.020)	(.177)	(.038)	(.018)
Mean of dependent variable	7.113	.126	.064	11.69	.825	.179

NOTE.—$N = 4{,}104$. Each estimate represents a separate regression using eq. (4) (both genders); (ln) income coefficients are adjusted using $\exp(\beta) - 1$.
* Significant at 10 percent.
** Significant at 5 percent.
*** Significant at 1 percent.

tical dashed lines contain at least one cohort exposed to the policy in utero. From figure 6 it is clear that the timing in the drop in relative outcomes for children of young mothers in the experimental regions is unusually large and fits very well with the number of weeks of in utero exposure to the policy. Higher exposure to the policy is strongly negatively linked to adult outcomes. The outcomes of the most exposed cohorts are substantially worse than of any other cohort during the observation period.[49]

3. Spillover Effects to Neighboring Regions

The APU speculated that the unexpectedly large increase in sales could be partly due to an increase in cross-border shopping from neighboring

[49] Appendix fig. A4 shows histograms of the estimated effects separately, and table A7 shows the separate estimates for all the key outcomes for the period November 1967 through December 1968. Figures A5 and A6 show that the results in fig. 6 are robust to excluding mothers above age 25 and mothers above age 21 (DD specification), respectively.

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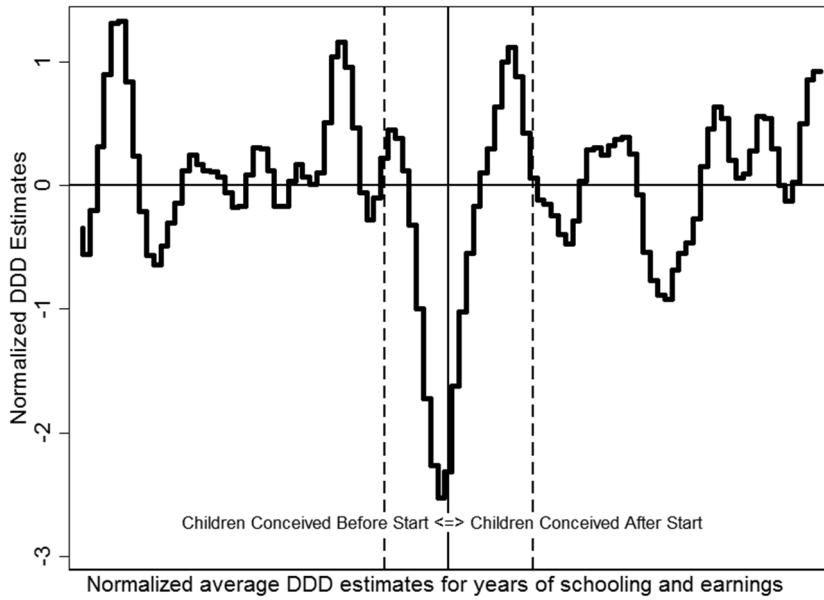


FIG. 6.—Normalized DDD estimates for years of schooling and earnings by cohort. The figure shows the locally weighted (0.05 bandwidth) average of the standardized β_3 estimates (mean zero, standard deviation one) for years of schooling and earnings (full sample) for all cohorts in the data. Specifically, I start by estimating equation (4) and set the EXPOSURE dummy equal to one for children born between January and April 1964 and then retain the estimated β_3. I then repeat the exercise after shifting the treatment window with increments of 1 month (i.e., February–May 1964, March–June 1964, etc.) until September–December 1972, that is, for all cohorts in the observation window (105 estimates per outcome). The estimates between the vertical dashed lines include at least one birth month cohort exposed to the policy experiment in utero.

regions. I next examine to what extent any such cross-border shopping also affected outcomes among children in the adjacent regions. Table 8 reports coefficients from the baseline specification after reassigning exposure to the cohort of children born between April and July 1968 to mothers under the age of 21 in one of the five regions neighboring the experiment area.[50] None of the coefficients are statistically significant. These results suggest that cross-border shopping, or other contemporaneous shocks affecting treatment and neighboring regions similarly, did not affect the outcomes of the children in the neighboring regions to any major extent. Since the neighboring regions and the treatment re-

[50] The regions are Kopparberg, Närke, Skaraborg, Älvsborg, and Halland. The experiment regions are excluded from these regressions.

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TABLE 8
IMPACT ON NEIGHBORING REGIONS: FULL SAMPLE

	DEPENDENT VARIABLE					
	Earnings	Zero Earnings	Welfare	Years of Schooling	High School Graduates	Higher Education
Exposed from 1st half of pregnancy	.047	.023	.005	−.071	−.006	−.015
	(.060)	(.017)	(.010)	(.117)	(.019)	(.027)
Observations	4,752	4,752	4,752	4,752	4,752	4,752

NOTE.—Each column represents a separate estimate using eq. (4). The outcomes are measured within each region of birth/year of birth/month of birth/mother < age 21 at delivery cell. All regressions are weighted by the number of children in each cell. Heteroscedasticity robust standard errors are reported in parentheses. Children born in the policy experiment regions are excluded. Results are similar if the sample is split by gender.

gions are highly economically interdependent, this provides additional support for the identifying assumptions of the main analysis.

VI. Extensions

A. Responsive Investments

1. Maternal Fixed-Effects Estimates

Another mediating mechanism suggested by the conceptual framework is postnatal responsive investments. Many studies have examined whether parents tend to accentuate, be neutral, or attenuate inequalities in their children's endowments by reallocating resources among children within the household in developing countries.[51] Conclusive evidence from developed countries, where equality concerns across siblings are likely to be more important and parents are less resource constrained, is still missing.[52] Parental responsive investments have important implications for the interpretation of studies interested in the biological impact of early-life conditions. In addition, if initial endowments are positively related to investments, policies that improve endowments could spur parents' willingness to invest, increasing the benefits of the intervention.

[51] Evidence of reinforcing strategies is found by Behrman, Rosenzweig, and Taubman (1994); Rosenzweig and Zhang (2009), Adhvaryu and Nyshadham (2014), and others find empirical support for a compensating strategy (Behrman, Pollak, and Taubman 1982; Pitt, Rosenzweig, and Hassan 1990). In China, Yi et al. (2015) document a reinforcing strategy for educational investments but compensatory for health investments.

[52] For example, following prenatal exposure to nuclear fallout, Almond et al. (2009) use the same approach and find evidence indicating *reinforcing* investments when considering school outcomes in Sweden, while Black et al. (2013) and Halla and Zweimuller (2014) find *compensating* investments in Norway and Austria, respectively.

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Almond and Currie (2010) suggest that the role of responsive investments can be assessed indirectly by examining the differences between the baseline DDD estimates and sibling fixed-effect estimates. Systematically larger (smaller) within-sibling estimates compared to the DDD estimates could suggest that parents may have chosen to reinforce (compensate) the differences in initial endowment induced by the policy.

Table 9 provides maternal fixed-effects (MFE) estimates showing that the exposed children have worse educational and labor market outcomes compared with their unexposed sibling.[53] The point estimates for educational outcomes are smaller than in the full sample, but it is hard to discern any systematic differences between the sibling sample DDD estimates (panel B) and the MFE estimates (panel A). The comparisons between the DDD and the MFE estimates provide indirect evidence indicating that parents are neutral. Alternatively, parents' responsive investment may have limited impact on the outcomes of their children, given the low information context, perhaps because they were made too late.[54] Most importantly, the robustness of the results to the inclusion of MFE is striking and provides strong support for the validity of the baseline model.

2. Fertility: Children and Grandchildren

Another salient way in which parents could respond to changes in initial endowments of their firstborn is to change desired family size. However, columns 1–3 of table 10 show that completed fertility of the mother was not affected.[55] Neither was the total number of grandchildren (irrespective of child exposure; cols. 4–6) or the number of children that the exposed child had before his or her 37th birthday (cols. 7–9). So despite the considerable impact on human capital and labor market outcomes, there are no adverse consequences for male or female fertility.[56]

[53] I retain the children whose sibling was born before 1977 (up to 5 years after the last baseline cohort) and reestimate the baseline model with individual-level data after adding MFE. Region effects are identified by between-birth movers.

[54] I also estimated separate MFE models by gender of the first child. The point estimates were larger for education for women in the MFE model than in the DDD model, whereas they were smaller for education in the MFE model than in the DDD model for males. For labor outcomes they were similar irrespective of model for both males and females. Taken at face value, this would indicate that parents compensated (reinforced) boys' (girls') educational investments but that this had little impact on their labor market outcomes. However, the precision of the estimates was not good enough to rule out identical estimates. See online app. table E1.

[55] I also checked if children exposed to the policy experiment were more likely to be the only child and, conditional on having a sibling, if they were more closely spaced. None of these specifications suggested that the policy altered the subsequent fertility decisions of the mother. These results are available on request.

[56] I also split the data along grandfather's income; the results were similar and showed no significant effects and no differences between high and low family income children. These results are available on request.

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TABLE 9
MATERNAL FIXED-EFFECTS ESTIMATES

	Dependent Variable: Labor and Education					
	Earnings	Zero Earnings	Welfare	Years of Schooling	High School Graduates	Higher Education
	A. DDD + Maternal Fixed Effects					
Exposed sibling	−.207**	.064***	.035**	−.200*	−.070**	−.018
	(.091)	(.020)	(.018)	(.120)	(.028)	(.023)
	B. Baseline DDD for the Sample with Siblings (Firstborns Only)					
Exposed sibling	−.279***	.046***	.031***	−.197***	−.055***	−.016
	(.075)	(.009)	(.010)	(.051)	(.010)	(.011)
Mean of dependent variable	7.106	.119	.0567	11.72	.833	.183
Number of sibships	203,772	246,164	246,164	246,164	246,164	246,164

NOTE.—Panel A reports maternal fixed-effects estimates in which the exposure variable is equal to one if one of the siblings was exposed to the experiment in utero and born to a mother under age 21. The control variables are the same as in eq. (4), but also maternal-specific indicators and variables that vary between the siblings (sex and month of birth indicators). For reference, panel B reports eq. (5) DDD estimates for the sibling sample.

* Significant at 10 percent.

** Significant at 5 percent.

*** Significant at 1 percent.

In summary, the absence of family size effects and the MFE estimates suggest that responsive parental investments throughout the child's life cycle do not seem to explain the effects on labor market outcomes. Without data on actual intrahousehold allocations, it is, however, difficult to draw strong conclusions based on these findings.

TABLE 10
REPRODUCTIVE OUTCOMES

	Mother's Reproductive Success						Children's Fertility by Age 37		
	Dependent Variable: Completed Family Size			Dependent Variable: Number of Grandchildren			Dependent Variable: Age 37 Family Size		
	Full Sample (1)	Men (2)	Women (3)	Full Sample (4)	Men (5)	Women (6)	Full Sample (7)	Men (8)	Women (9)
Exposed from 1st half of pregnancy	.074	.035	.111	.174	.078	.287	.027	.021	.036
	(.066)	(.098)	(.068)	(.111)	(.134)	(.227)	(.054)	(.092)	(.088)
Observations	4,104	4,104	4,104	4,104	4,104	4,104	4,104	4,104	4,104
Mean of dependent variable (young mothers)	2.5	2.5	2.5	3.615	3.399	3.843	1.730	1.516	1.954

NOTE.—Each column represents a separate regression using eq. (4).

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B. *Has the Policy Experiment Affected Health in the Subsequent Generation?*

Many studies focus on effects of in utero alcohol exposure in the exposed generation (F1), but the evidence on intergenerational effects is restricted to animal experiments. In general, the evidence on intergenerational effects in humans from in utero shocks is mixed and stems primarily from studies on prenatal exposure to the severe malnutrition following the Dutch Hunger Winter.[57] However, the evidence from the Dutch Hunger Winter on fertility is mixed.[58] Without evidence on selection into fertility in the F1 generation, the interpretation of any impact on the next generation (F2) is unclear.

Table 10 showed that selection into fertility (among the F1s) was not affected. It is therefore straightforward to examine the impact on the F2s without worrying about sample selection effects. Table 11 provides results for health at birth in the F2s. There are no clear effects from either maternal or paternal prenatal exposure on the share prematurely born, low birth weight, or share sons. To my knowledge this is the first test of transgenerational effects of parental prenatal alcohol exposure in humans.

VII. Summary and Conclusions

I examine the long-run effects of in utero exposure to a temporary "liberalization" of alcohol sales following an alcohol policy experiment in two Swedish regions in the late 1960s, before the risks associated with alcohol consumption during pregnancy were widely recognized and well before women were recommended to abstain from alcohol during pregnancy. I find that prenatal exposure to the policy is linked to substantially worse labor market outcomes later in life.

I also find that males are more affected than females as reflected by a lower sex ratio and systematically larger long-term effects on males. The relative imperturbability of girls calls for further research on the role of prenatal and the early environment on gender differences in later outcomes, for example, the still poorly understood and astounding gender gap in old-age longevity (Cullen, Cummins, and Fuchs 2012; Cullen

[57] See Lumey (1992), Stein and Lumey (2000), and Veenendaal et al. (2013). Other transgenerational studies are Almond et al. (2010) and Richter and Robling (2013) on the Chinese famine and the Spanish flu, respectively. Fertility and paternity effects are not examined in the former, and the latter study does not assess fertility selection effects.

[58] Painter et al. (2008) find positive effects, while Lumey and Stein (1997) find no impact on fertility.

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TABLE 11
INTERGENERATIONAL EFFECTS

	(1)	(2)	(3)	(4)	(5)	(6)
Exposed parent (F1)	Mother	Mother	Mother	Father	Father	Father
Birth outcome (F2)	Low birth weight	Premature	Share males	Low birth weight	Premature	Share males
Parent exposed from 1st half of pregnancy	−.0099	.0046	.0052	−.0068	−.0109	−.0203
	(.0219)	(.0332)	(.0238)	(.0159)	(.0316)	(.0421)
Observations	4,104	4,104	4,104	4,104	4,104	4,104
Mean of outcome variable	.051	.073	.513	.040	.063	.515

NOTE.—Columns 1 and 4 report the impact of maternal prenatal exposure to the policy on the share of exposed mothers whose first child weighed < 2,500 grams at birth. Columns 2 and 5 report estimates for the share of firstborn children born before 37 weeks' gestation. Columns 3 and 6 report estimates for the impact on the share of sons. Parent exposed in utero is a dummy variable taking value one if the mother or the father was exposed to the policy from early pregnancy, and zero otherwise. The table reports estimates from eq. (4). Birth outcomes come from the Medical Birth Register and cover all children born in Sweden from 1973 through 2009.

et al. 2015) or the gender gap in behavioral, educational, and later outcomes (e.g., Kessler et al. 2006; Chetty et al. 2016).

However, changes in adult outcomes did not affect fertility in the exposed (F1) cohort, nor health at birth in the subsequent generation (F2). The absence of intergenerational effects potentially reflects the relatively muted impact of the policy on disposable income. Future work could exploit the policy to gain insights into the role of epigenetic effects from prenatal alcohol exposure by examining, for example, the development of alcohol-related diseases in the F1 and F2 cohorts. If such effects could be documented, they would have strong implications for our understanding of the vulnerability to alcoholism. Since assignment of exposure is readily observed in available register data, future multigenerational studies can be conducted with minimal sample attrition.

Data limitations prohibit conclusions about how the policy affected alcohol consumption patterns. For example, it is unclear whether it generated an increase in daily drinking or whether it increased binge drinking during weekends. Hence, conclusions on whether the estimated impact of the policy is due to some specific consumption pattern or another is not possible. It is, however, well established that the almost exclusive consumption pattern among young people in Sweden is binge drinking during weekends and festivities.[59] In the United States, about 90 percent of

[59] The pattern of drinking in Sweden has been characterized by nondaily drinking, irregular binge drinking episodes (e.g., during weekends and at festivities), and the acceptance of drunkenness in public; see, e.g., Kühlhorn et al. (1999).

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the alcohol consumed by youths is in the form of binge drinks (Office of Juvenile Justice and Delinquency Prevention 2001). Mosher, Jones, and Abma (2012) estimate that 80 percent (50 percent) of teenage (all) pregnancies are unplanned. Combined with the ubiquity of alcohol consumption before pregnancy recognition, the results in this study clearly call for more research on the effects of prenatal alcohol exposure on child outcomes.[60] The substantial impact of the policy suggests that effective policy measures that reduce drinking among young adults may not only reduce individual alcohol-related morbidity and mortality (Carpenter and Dobkin 2009) but potentially also improve long-term outcomes of their children.

Appendix

Figures and Tables

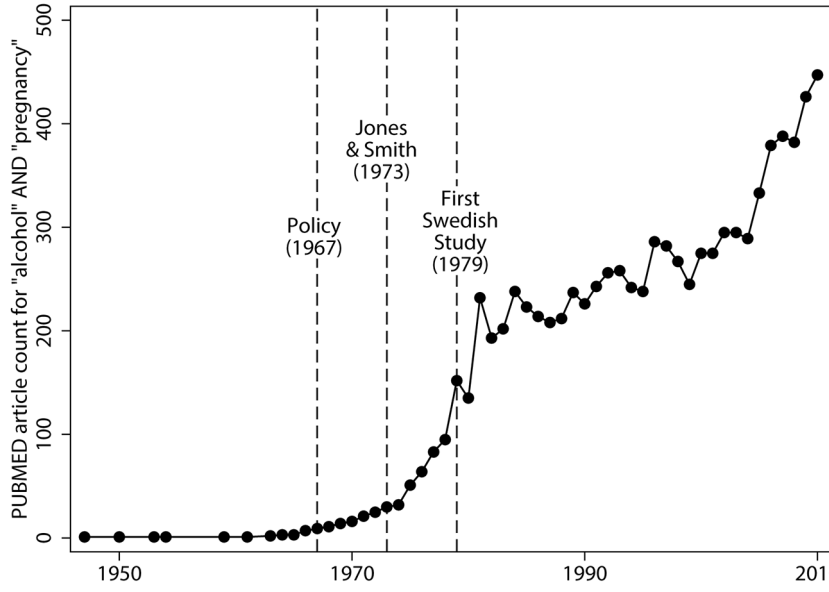


FIG. A1.—PUBMED article search for "alcohol" and "pregnancy"

[60] See online app. B for a review of the prevalence of alcohol consumption during pregnancy in several countries.

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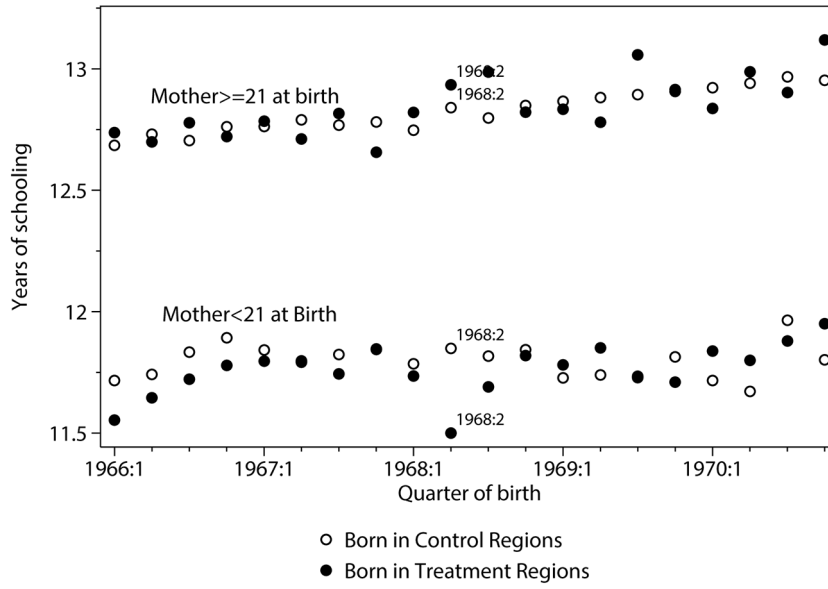


FIG. A2.—Average years of schooling 2006, treated versus control, by age of mother at birth (full sample).

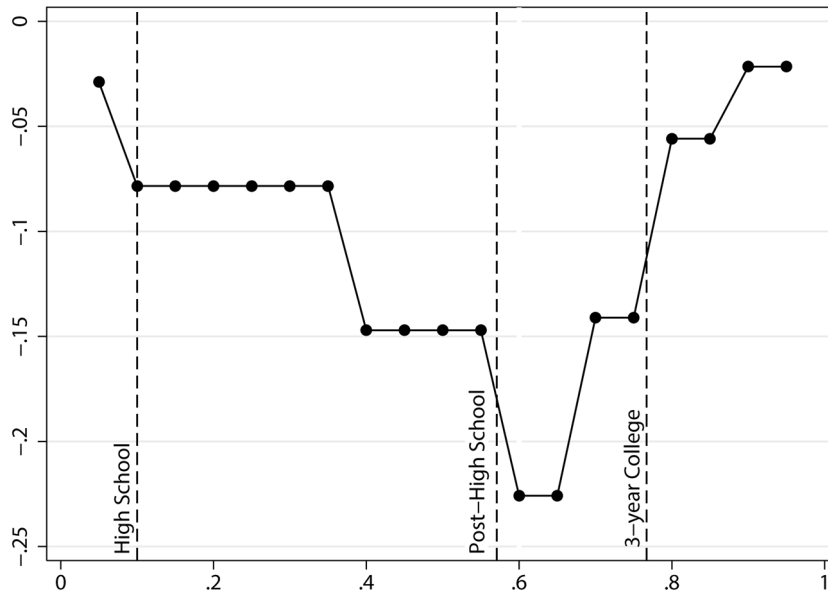


FIG. A3.—Unconditional quantile estimates for years of schooling. The figure reports unconditional quantile regression estimates (Firpo et al. 2009) for the full sample.

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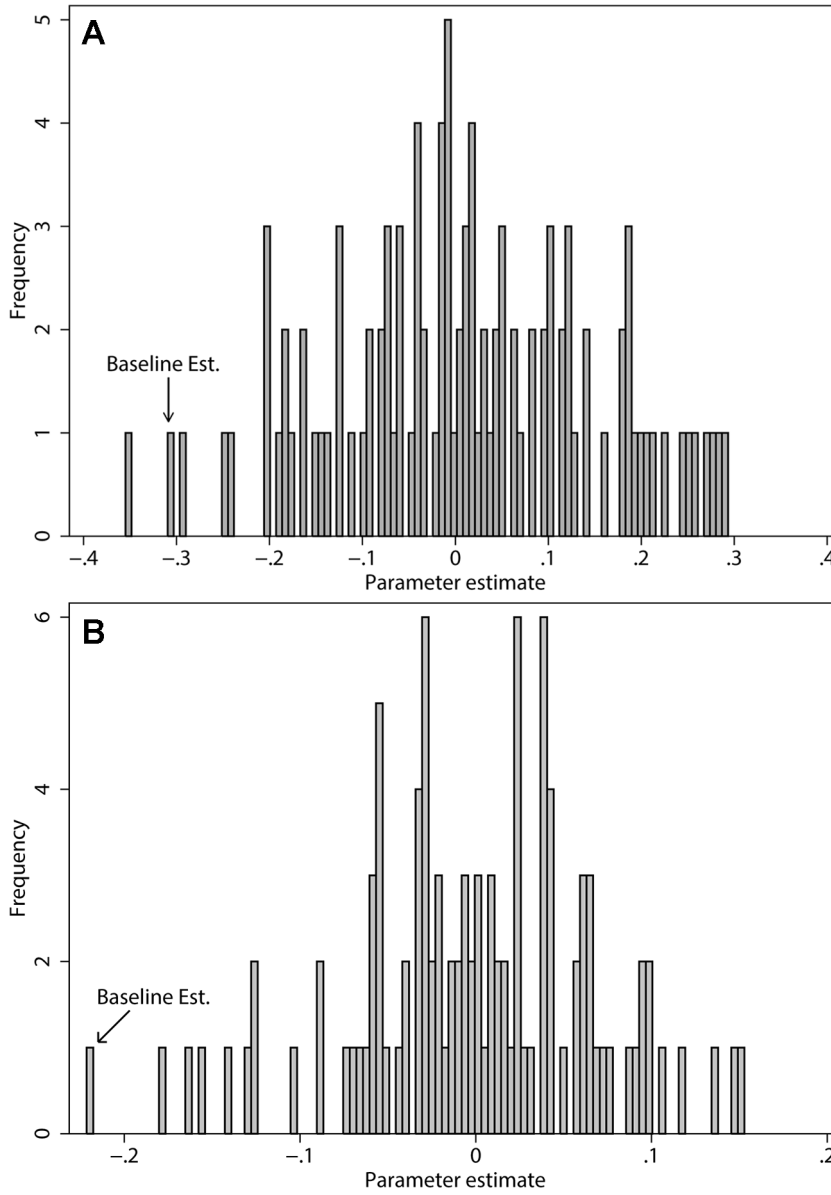


FIG. A4.—Placebo estimates. The figure shows the estimate of equation (5) for the full sample using a rolling window approach from 1964 through 1972 assigning an exposure window around a false policy exposure date. There are 105 estimates for each figure in which each estimate increases the false exposure date by 1 month. Note that the placebo estimates are not independent of each other, as the samples overlap.

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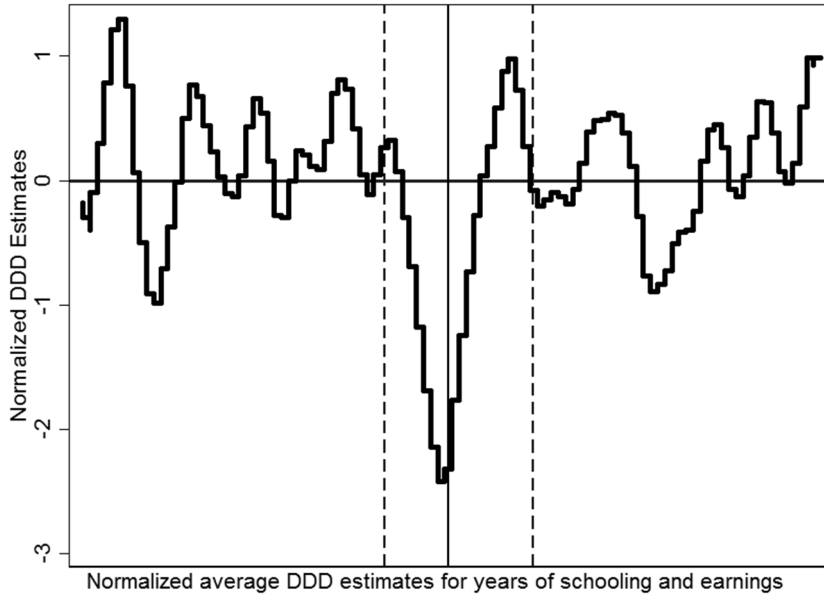


FIG. A5.—Normalized average DDD estimates for years of schooling and earnings (mothers under age 25 only). See the text and figure 6 for more details on how this figure is constructed.

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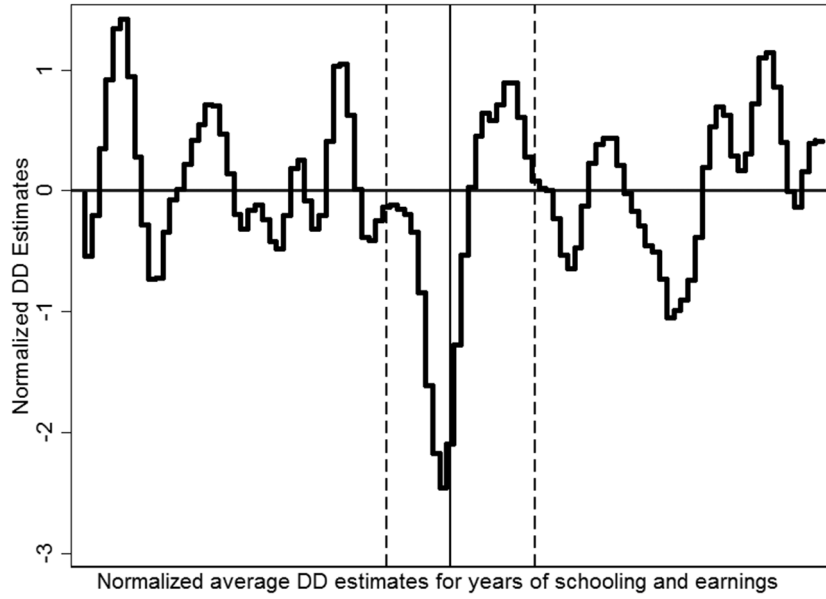


FIG. A6.—Normalized average DD estimates for years of schooling and earnings (January–April 1964 through September–December 1972), using young mothers only. See the text and figure 6 for more details on how this figure is constructed.

TABLE A1
IMPACT OF THE POLICY ON CONTEMPORARY ARRESTS FOR DRUNKENNESS

	A. Timing					
	(1)	(2)	(3)	(4)	(5)	(6)
Treated×1(policy)	.0964***		.1105***	.0859***	.0905**	
	(.0259)		(.0337)	(.0262)	(.0358)	
Treated×1 (Q4:1967)[a]		.1010*				.0902*
		(.0570)				(.0474)
Treated×1 (Q1:1968)		.1368**				.1321
		(.0493)				(.0813)
Treated×1 (Q2:1968)		.0492				.0562
		(.0450)				(.0416)
Treated×1(policy ($t-1$))			.0474			
			(.0309)			
Treated×1(policy ($t+1$))			.0018			
			(.0449)			
Neighbor×1 (policy)				−.0466		
				(.0695)		
Observations	672	672	672	672	672	672

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TABLE A1 (*Continued*)

	A. Timing					
	(1)	(2)	(3)	(4)	(5)	(6)
Region×year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Treatment region×quarterly trends					Yes	Yes
	B. Population					
	Women < 21 (1)	Men < 21 (2)	Women ≥ 21 (3)	Men ≥ 21 (4)	All Women (5)	All Men (6)
Treated×1(policy)	.1952	.1337	−.0037	−.0040		
	(.1442)	(.0747)	(.0853)	(.0385)		
Treated×1(<21)×1(policy)					.1989**	.1377**
					(.0875)	(.0562)
Observations	20	20	20	20	40	40
City fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
City×year fixed effects					Yes	Yes

Source.—Statistics Sweden 1964–70 (panel A) and 1968 (panel B).

Note.—Panel A reports region-level DD regression of policy on the log of the number of arrests for drunkenness (all ages). The unit of observation is region/quarter for the period 1964–70. The original data are at the city-quarter level. To construct the region-level data, city-level log arrests are regressed on city-specific effects, and the residuals are then aggregated to the region level (the level of treatment). Results are highly similar if city-level data are used instead. Robust standard errors (in parentheses) are clustered on the region. All specifications controls for year-quarter time effects. Baseline regression in col. 1 controls for region-year fixed effects and shows the average effect of the policy on arrests. Column 2 shows the impact of the policy across the differing quarters during which the policy was active. Panel B reports gender-specific estimates using yearly data on (log) arrests from Stockholm, Malmö, Göteborg, and the rest of the large cities (i.e., in total four observations per year), which are consistently reported for the period 1964–68. Because of the yearly data, the policy period is defined as one if $t = \{1967, 1968\}$, and zero otherwise. Columns 1–4 report DD estimates for males and females above and below age 21. Columns 5 and 6 report DDD estimates separately for females and males for those under age 21 relative to those over 21. Robust standard errors are in parentheses.

[a] Note that Q4:1967 includes arrests made in October 1967 when the policy had not yet been implemented. Assuming that the effect in October 1967 is not different from zero, the Q4:1967 estimate in col. 2 implies that the average effect for the first 2 months of the experiment in the number of arrests increased by 15 percent ($[0.1 \times 3]/2$). Columns 3 and 4 are placebo specifications, showing that the effects are not likely to be driven by potential differential trends that the region-year fixed effects do not capture. There are no significant differences in the treatment and control regions in the year before (i.e., Q4:1966–Q2:1967) or after the policy started (i.e., Q4:1968–Q2:1969), nor in the in the five neighboring regions at the same time as the policy (col. 4). Columns 5 and 6 replicate cols. 1 and 2, adding treatment region-specific quadratic quarter trends to account for differences in seasonal patterns in arrest rates across the treatment and control regions.

* Significant at 10 percent.

** Significant at 5 percent.

*** Significant at 1 percent.

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TABLE A2
Impact on Early-Life Health

	Difference-in-Differences: Young Mothers Sample						Difference-in-Difference-in-Differences: All Mothers Sample					
Policy Exposure Group	Share of Males (1)	Male Cohort Size (2)	Female Cohort Size (3)	Share of Males (4)	Month of Birth Males (5)	Month of Birth Females (6)	Share of Males (7)	Male Cohort Size (8)	Female Cohort Size (9)	Share of Males (10)	Month of Birth Males (11)	Month of Birth Females (12)
Exposed in utero					−.224***	.014					−.28*	.056
					(.072)	(.095)					(.134)	(.141)
Born just before policy started	.002	−.013	.016	.002			.010	.0141	−.050	.011		
	(.018)	(.051)	(.072)	(.018)			(.020)	(.0634)	(.087)	(.025)		
Exposed from 2nd half of pregnancy	−.007	−.056	−.071				−.004	−.0929	−.108			
	(.019)	(.069)	(.071)				(.0272)	(.0833)	(.088)			
Exposed from 1st half of pregnancy	−.062**	−.203**	.049				−.073**	−.176*	.093			
	(.028)	(.092)	(.077)				(.0309)	(.1158)	(.078)			
Exposed from 3rd trimester				.008						.008		
				(.023)						(.041)		
Exposed from 2nd trimester				−.032						−.029		
				(.030)						(.029)		

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Exposed from 1st trimester				−.070***						−.08***		
				(.022)						(.026)		
Conceived during policy	.009	−.012	−.048	.009			.023	.0053	−.048	.023		
	(.016)	(.053)	(.061)	(.016)			(.0273)	(.0711)	(.082)	(.027)		
Conceived just after policy ended	.012	.062	−.032	.012			.031	.1336	−.076	.044		
	(.027)	(.065)	(.147)	(.027)			(.030)	(.0944)	(.173)	(.0354)		
Observations	2,052	2,052	2,052	2,052	171	171	4,104	4,104	4,104	4,104	342	342
Mean of dependent variable (monthly cells)	.512	22	21	.512	4.05	4.05	.514	44	41	.514	4.05	4.05

NOTE.—Each column represents a separate regression using monthly data. Born just before policy started/conceived just after policy ended are defined as being born within 6 months of the start of the policy started and estimated to have been conceived within 6 months after the policy ended. For the log cohort size, coefficients are adjusted using $\exp(\beta) - 1$. Columns 1–3 and 7–9 report estimates from the model in eq. (5). Columns 4 and 10 report extended versions of eq. (5) allowing the effects of the policy to vary by estimated gestational trimester at the start of the policy experiment. Columns 5, 6, 11, and 12 report estimates from a specification with the month of birth as the dependent variable only for the sample born before August each year and replace the year × month effects with year effects.

* Significant at 10 percent.

** Significant at 5 percent.

*** Significant at 1 percent.

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TABLE A3
EFFECTS ON EARLY-LIFE HEALTH AND SAMPLE COMPOSITION

	Dependent Variable									
	Grandfather Income 1968	Share of Males				Grandparent High School 1990	Share of Males			
Policy Exposure Group	Full Sample (1)	Grandfather Income 1968 Sample (2)	Grandfather Income 1968 Sample (3)	Grandfather Income High 1968 Sample (4)	Grandfather Income Low 1968 Sample (5)	Young Sample (6)	Grandparent Education Sample (7)	Grandparent Education Sample (8)	Grandparent ≥ High School Sample (9)	Grandparent <High School Sample (10)
Born before policy started	.019	.013	.013	.015	.017	.0008	−.0137	−.0137	−.0434	−.0035
	(.031)	(.024)	(.024)	(.035)	(.037)	(.0302)	(.0283)	(.0283)	(.0644)	(.0279)
Conceived before, exposed from 2nd half of pregnancy	−.010	.001		.003	−.014	.0414	−.0415		−.0619	−.0341
	(.035)	(.026)		(.044)	(.047)	(.0359)	(.0335)		(.0459)	(.0418)
Conceived before, exposed from 1st half of pregnancy	.003	−.076**		−.059	−.090**	.0393	−.105***		−.1047	−.108***
	(.032)	(.035)		(.044)	(.038)	(.0354)	(.0323)		(.0817)	(.0270)
Exposed from 3rd trimester			.007					−.0577		
			(.036)					(.0459)		

Exposed from 2nd trimester			−.026					−.0372		
			(.033)					(.0426)		
Exposed from 1st trimester			−.081**					−.115***		
			(.034)					(.0214)		
Conceived during policy	−.061**	.027	.027	.013	.049	−.0351	.0083	.0083	.0641	−.0135
	(.029)	(.028)	(.028)	(.035)	(.036)	(.0276)	(.0216)	(.0216)	(.0459)	(.0256)
Conceived after policy ended	−.026	.028	.028	−.010	.064	.0234	.0080	.0080	.0064	.0085
	(.037)	(.041)	(.041)	(.054)	(.044)	(.0288)	(.0353)	(.0353)	(.0564)	(.0382)
Observations	4,104	4,104	4,104	4,104	4,104	2,037	2,037	2,037	1,830	2,026
Mean of dependent variable (young mothers)	9.98	.513	.513	.514	.513	.303	.517	.517	.513	.513

NOTE.—Each column represents a separate regression using monthly data. Columns 1–2, 4–5, 6–7, and 9–10 report estimates from the model in eq. (5). Columns 3 and 8 report extended versions of eq. (5) allowing the effects of the policy to vary by estimated trimester at the start of the policy experiment. Grandparents' educational attainments are available for those whose grandparents were born after 1925 and were alive in 1990 (i.e., 65 and under in 1990). In practice it limits the sample to the under age 21 women who themselves where born to women under age 20 at birth.

* Significant at 10 percent.

** Significant at 5 percent.

*** Significant at 1 percent.

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TABLE A4
EFFECTS ON FAMILY COMPOSITION

	Dependent Variable					
	Grandfather's Income 1968			Any Younger Half Siblings?		
Policy Exposure Group	All (1)	Male (2)	Female (3)	All (4)	Male (5)	Female (6)
Born before policy started	.0193	.0368	.0053	−.0276*	−.0335	−.0212
	(.0312)	(.0363)	(.0551)	(.0165)	(.0288)	(.0263)
Conceived before, exposure from 2nd half of pregnancy	−.0101	−.0433	.0127	.0060	.0329	−.0256
	(.0354)	(.0679)	(.0496)	(.0276)	(.0458)	(.0353)
Conceived before, exposure from 1st half of pregnancy	.0030	.0133	−.0099	−.0290	−.0427	−.0249
	(.0316)	(.0412)	(.0443)	(.0361)	(.0386)	(.0509)
Conceived during policy:						
Early	−.0836**	−.0999**	−.0686	.0573***	.0514*	.0585**
	(.0393)	(.0421)	(.0627)	(.0207)	(.0286)	(.0238)
Late	−.0303	−.0542	−.0032	−.0155	−0.0106	−.0557
	(.0364)	(.0487)	(.0719)	(.0224)	(0.0451)	(.0406)
Conceived after policy ended	−.0256	−.0728	.0243	−.0315	0.0126	−.0811*
	(.0373)	(.0513)	(.0589)	(.0312)	(0.0341)	(.0438)
Observations	4,104	4,104	4,104	4,104	4,104	4,104
Mean of dependent variable (young mothers)	9.98	9.98	9.98	.36	.36	.36

NOTE.—Each column represents a separate regression using monthly data. Grandfather's income is measured in 1968 (median age 55). Family instability is proxied by whether the focal child has any younger half siblings (see the text for details). The table reports estimates from an extended version of eq. (5) allowing for differential effects depending on when during the policy the child was conceived early (November 1967–February 1968) or late (March 1968–July 1968). The timing of conception is estimated using the month of birth.

* Significant at 10 percent.

** Significant at 5 percent.

*** Significant at 1 percent.

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TABLE A5
IMPACT ON LABOR MARKET OUTCOMES

Policy Exposure Group	Earnings			Zero Earnings			Welfare		
	All (1)	Male (2)	Female (3)	All (4)	Male (5)	Female (6)	All (7)	Male (8)	Female (9)
	A. Difference-in-Difference-in-Differences								
Born just before policy started	.017	−.032	.060	−.008	.003	−.020	.006	.002	.011
	(.048)	(.067)	(.084)	(.016)	(.022)	(.022)	(.012)	(.013)	(.017)
Exposed from 2nd half of pregnancy	−.011	−.003	−.031	.004	−.002	.013	−.011	−.014	−.007
	(.067)	(.058)	(.114)	(.020)	(.016)	(.033)	(.012)	(.017)	(.012)
Exposed from 1st half of pregnancy	−.242***	−.2442**	−.1553	.073***	.083***	.060	.035***	.044**	.029**
	(.090)	(.113)	(.139)	(.027)	(.024)	(.043)	(.013)	(.022)	(.013)
Conceived during policy	.017	.039	−.043	−.005	−.012	.004	−.003	−.020*	.016
	(.065)	(.070)	(.077)	(.012)	(.017)	(.020)	(.009)	(.012)	(.012)
Conceived just after policy ended	.043	−.101	.151	.022	.024	.025	.007	.014	−.001
	(.053)	(.071)	(.102)	(.019)	(.025)	(.030)	(.011)	(.020)	(.017)
Observations	4,104	4,104	4,104	4,104	4,104	4,104	4,104	4,104	4,104
Mean of dependent variable	7.133	7.440	6.801	.062	.057	.068	.123	.112	.135
	B. Difference-in-Differences								
Born just before policy started	−.016	−.014	−.021	−.010	−.010	−.011	.006	.004	.009
	(.038)	(.052)	(.069)	(.010)	(.016)	(.014)	(.012)	(.011)	(.015)
Exposed from 2nd half of pregnancy	−.047	−.052	−.053	−.004	−.017	.010	−.015	−.017	−.013
	(.045)	(.050)	(.082)	(.016)	(.014)	(.025)	(.010)	(.015)	(.011)
Exposed from 1st half of pregnancy	−.222***	−.211**	−.159	.051**	.053**	.043	.022**	.033*	.013
	(.078)	(.095)	(.120)	(.023)	(.021)	(.036)	(.010)	(.018)	(.009)
Conceived during policy	.041	.070	−.013	−.002	−.007	.003	−.002	−.013	.011
	(.047)	(.051)	(.061)	(.008)	(.012)	(.016)	(.007)	(.009)	(.010)
Conceived just after policy ended	−.012	−.100*	.069	.020	.021	.020	.009	.021	−.004
	(.044)	(.057)	(.082)	(.015)	(.021)	(.021)	(.010)	(.016)	(.012)
Observations	4,104	4,104	4,104	4,104	4,104	4,104	4,104	4,104	4,104
Mean of dependent variable	7.133	7.440	6.801	.123	.112	.135	.062	.057	.068

NOTE.—Each column represents a separate regression using monthly data. The earnings coefficients are adjusted using $\exp(\beta) - 1$. For panel A, the estimated model is described in eq. (5). Panel B reports results from a DD version of eq. (5), where only children of young mothers have been retained. This simpler model accounts for only region and year × month effects.

* Significant at 10 percent.
** Significant at 5 percent.
*** Significant at 1 percent.

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TABLE A6
EFFECTS ON COGNITIVE AND NONCOGNITIVE ABILITY SUBSCORES

	Outcome Variable			
	Average Score	Low Score	Medium Score	High Score
	A. Cognitive Ability			
i. Fluid intelligence test: logical ability:				
In utero (1–4)	−.071	.085***	−.062**	−.023
	(.064)	(.030)	(.027)	(.025)
R^2	.82	.65	.16	.78
Mean of outcome variable	−.177	.271	.562	.167
ii. Fluid intelligence test: spatial ability:				
In utero (1–4)	−.055	.048**	−.024	−.024
	(.056)	(.022)	(.026)	(.035)
R^2	.76	.58	.08	.67
Mean of outcome variable	−.158	.259	.555	.187
iii. Crystallized intelligence test: synonyms:				
In utero (1–4)	−.116***	.041*	−.030	−.011
	(.044)	(.023)	(.029)	(.018)
R^2	.84	.70	.27	.82
Mean of outcome variable	−.196	.274	.609	.117
iv. Crystallized intelligence test: technical comprehension:				
In utero (1–4)	.110*	−.038*	.018	.020
	(.059)	(.021)	(.058)	(.043)
R^2	.78	.57	.15	.75
Mean of outcome variable	−.175	.258	.593	.149
	B. Noncognitive Ability			
i. Social maturity:				
In utero (1–4)	−.120	.056***	−.043	−.014
	(.105)	(.014)	(.065)	(.069)
R^2	.61	.42	.32	.59
Mean of outcome variable	−.125	.185	.577	.238
ii. Intensity:				
In utero (1–4)	−.109**	.030	.025	−.055
	(.049)	(.026)	(.064)	(.052)
R^2	.51	.43	.17	.47
Mean of outcome variable	−.0384	.320	.434	.246
iii. Psychological energy:				
In utero (1–4)	.052	.004	−.054	.050
	(.088)	(.018)	(.041)	(.043)
R^2	.57	.34	.31	.56
Mean of outcome variable	−.125	.163	.614	.223

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TABLE A6 (*Continued*)

	Outcome Variable			
	Average Score	Low Score	Medium Score	High Score
	B. Noncognitive Ability			
iv. Emotional stability:				
In utero (1–4)	−.016	.005	.006	−.011
	(.077)	(.020)	(.047)	(.051)
R^2	.58	.42	.56	.68
Mean of outcome variable	−.117	.188	.656	.156

Note.—The table reports estimates from eq. (4) on the effects on noncognitive skills and effects on subscores of the noncognitive skills score as graded by trained psychologists at military enlistment procedure at age 18 for males. For ease of comparison, all scores have been standardized, with mean zero and standard deviation one. The share low, medium, and high scores are defined on the basis of the original scores, which take on integer values between 1 and 5, with 5 being the highest score: low, 1–2; medium, 3; and high, 4–5. The scores are standardized within each enlistment year cohort. See the text for explanations of the different subscores.

* Significant at 10 percent.

** Significant at 5 percent.

*** Significant at 1 percent.

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TABLE A7

IMPACT OF THE EXPERIMENT DEPENDING ON GESTATIONAL AGE AT THE START OF THE EXPERIMENT: DEPENDENT VARIABLES: EDUCATIONAL, LABOR MARKET, AND HEALTH-RELATED OUTCOMES

	PERIOD OF BIRTH (Estimated Gestational Age in Months)										
OUTCOME	Nov.–Feb. (6–9 Mos.) (1)	Dec.–March (5–8 Mos.) (2)	Jan.–April (4–7 Mos.) (3)	Feb.–May (3–6 Mos.) (4)	March–June (2–5 Mos.) (5)	April–July (1–4 Mos.) (6)	May–Aug. (NC–3 Mos.) (7)	June–Sept. (NC–2 Mos.) (8)	July–Oct. (NC–1 Mo.) (9)	Aug.–Nov. (NC) (10)	Sept.–Dec. (NC) (11)
Labor earnings	−.019	.008	−.054	−.160	−.190*	−.244***	−.184*	−.075	−.035	.006	.007
	(.040)	(.080)	(.095)	(.111)	(.110)	(.090)	(.108)	(.067)	(.070)	(.073)	(.075)
Zero earnings	−.009	.018	.055*	.072***	.076***	.072***	.032**	.012	−.004	−.009	−.028*
	(.020)	(.019)	(.029)	(.025)	(.026)	(.027)	(.016)	(.022)	(.020)	(.017)	(.014)
Welfare dependent	−.013	−.008	.002	.004	.025*	.035***	.028**	.021	.005	−.002	−.006
	(.014)	(.014)	(.015)	(.013)	(.015)	(.012)	(.013)	(.016)	(.012)	(.012)	(.012)
Years of schooling	.016	−.040	−.200**	−.244**	−.250**	−.309***	−.355***	−.295**	−.163	−.200	.019
	(.118)	(.093)	(.101)	(.100)	(.108)	(.089)	(.106)	(.128)	(.147)	(.141)	(.123)
High school graduate	−.012	−.013	−.040**	−.052***	−.063***	−.063***	−.052***	−.024	−.019	−.021	−.002
	(.014)	(.016)	(.019)	(.018)	(.016)	(.016)	(.013)	(.016)	(.017)	(.017)	(.018)
University graduate	.016	.003	−.033	−.025	−.019	−.032*	−.049**	−.058**	−.029	−.039	.000
	(.030)	(.025)	(.023)	(.022)	(.022)	(.018)	(.023)	(.025)	(.035)	(.034)	(.031)
Share males	−.003	.000	−.011	−.061**	−.067**	−.075**	−.045	−.031	−.019	−.002	.002
	(.029)	(.026)	(.030)	(.028)	(.029)	(.030)	(.033)	(.032)	(.040)	(.043)	(.045)
Observations	4,104	4,104	4,104	4,104	4,104	4,104	4,104	4,104	4,104	4,104	4,104

NOTE.—Each column represents a separate regression using the model in eq. (4). The outcomes are averages/fractions within each region of birth/month of birth/mother < age 21 at delivery cell. All regressions are weighted by the cell size used to calculate the dependent variable. Heteroscedasticity robust standard errors are reported in parentheses. The estimates from using the original treatment window are reported in col. 6. NC = no one conceived.

* Significant at 10 percent.

** Significant at 5 percent.

*** Significant at 1 percent.

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